

IoT Mid-Term Exam — Section 1 Complete Answers

Silicon University | BTCS-T-PC-056 | All Papers (2023–2026)

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1a (MCQ)	Answer: 4. Message Queue Telemetry Transport — MQTT stands for Message Queue Telemetry Transport. It is a lightweight publish-subscribe messaging protocol designed for constrained devices and low-bandwidth, high-latency networks.
1b	Role of Controller Service in IoT: The controller service acts as the brain of an IoT system. It receives data from sensors/devices, processes it, makes decisions based on predefined logic or rules, and sends commands to actuators. It manages device communication, event handling, and coordination between different components of the system.
1c	WebSocket APIs use the Request-Response communication model, but more specifically they enable full-duplex, bidirectional communication. Unlike traditional HTTP, WebSocket keeps the connection open, allowing both client and server to send data at any time, enabling real-time data exchange.
1d	Network Layer Protocols in IoT Protocol Suite: IPv4, IPv6, 6LoWPAN (IPv6 over Low-Power Wireless Personal Area Networks), RPL (Routing Protocol for Low-Power and Lossy Networks).
1e	DHT11 and DHT22 sensors are used for humidity and environment temperature measurement. SHT31 and BME280 are also commonly used for combined humidity and temperature sensing in IoT applications.
1f	Smart Farming: Smart farming (also called precision agriculture) refers to the use of IoT technologies, sensors, GPS, drones, big data, and AI to monitor and optimize agricultural processes such as crop growth, soil conditions, irrigation, and livestock management. It aims to increase yield, reduce waste, and improve sustainability.

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1a	Embedded System vs IoT: An embedded system is a dedicated computer system designed to perform specific tasks within a larger system (e.g., a microcontroller in a washing machine). It typically operates standalone without internet connectivity. IoT (Internet of Things) extends embedded systems by adding internet/network connectivity, enabling devices to collect, exchange, and act on data remotely. IoT devices are interconnected and can be monitored/controlled over the internet.
1b	Role of Controller Service in IoT: The controller service processes sensor data, applies business logic/rules, and sends control commands to actuators. It manages the flow between sensing and actuation, handles event-driven responses, and often communicates with cloud services or local gateways.
1c	Transmission Range and Data Rate of a Sensor Node: Transmission range refers to the maximum distance over which a sensor node can reliably communicate (e.g., Zigbee: ~10-100m, LoRa: up to 15 km). Data rate is the speed at which data is transmitted (e.g., Zigbee: 250 kbps, Bluetooth: 1 Mbps, LoRa: 0.3–50 kbps). Both parameters determine the suitability of a sensor node for a specific IoT application.
1d	Idle Listening in Communication: Idle listening occurs when a node keeps its radio receiver turned ON even when no data is being transmitted. It wastes significant

	energy in wireless sensor networks (WSNs). MAC protocols like S-MAC use sleep schedules to minimize idle listening and conserve battery.
1e	Drawback of IP Protocol: The IP protocol is connectionless and does not guarantee delivery, ordering, or error correction. Traditional IPv4 has a limited address space (4.3 billion addresses), which is insufficient for the billions of IoT devices. IPv6 was introduced to address this, but transition overhead and complexity remain challenges.
1f	Wearable Electronics: Wearable electronics are smart devices worn on the body (e.g., smartwatches, fitness bands, smart glasses, health monitors). They are used for health monitoring (heart rate, SpO2, steps), activity tracking, navigation, communication, and augmented reality applications.

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1a (MCQ)	Answer: 3. Transport Layer — The Transport layer (e.g., TCP, UDP) provides end-to-end communication in IoT. It ensures data is delivered reliably or in a timely manner between source and destination devices across the network.
1b	Analytic Component in IoT: The analytic component in IoT refers to the software/hardware layer responsible for processing, analyzing, and deriving insights from raw data collected by IoT devices. It includes data analytics, machine learning models, real-time stream processing, and visualization tools that help make intelligent decisions from sensor data.
1c	Cacheable and Non-Cacheable Constraints (REST): In REST architecture, cacheable means responses can be stored by clients or intermediaries to improve efficiency and reduce server load (e.g., GET responses with cache headers). Non-cacheable means responses must not be stored and require fresh data from the server every time (e.g., real-time sensor readings that change constantly).
1d	Role of IoT in Machine Diagnosis and Prognosis: IoT sensors continuously monitor machine parameters (temperature, vibration, pressure, RPM). This data is analyzed to diagnose current faults and predict future failures (predictive maintenance). It helps reduce unplanned downtime, extend machine life, and lower maintenance costs in industrial settings.
1e	Functionalities of AirPi: AirPi is a Raspberry Pi-based IoT air quality and weather monitoring kit. Its functionalities include measuring temperature, humidity, air pressure, carbon monoxide (CO) levels, nitrogen dioxide (NO2), and light levels. It uploads data to the internet for remote monitoring and environmental analysis.
1f	Use of LoRa in Agriculture: LoRa (Long Range) is used in agriculture for long-distance, low-power wireless communication. Applications include: soil moisture monitoring, weather station data transmission, livestock tracking, irrigation control systems, and crop monitoring over large farm areas where cellular coverage may be unavailable.

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1a	Embedded System vs IoT: An embedded system is a specialized computing system designed for dedicated tasks within a device, typically without internet connectivity. IoT devices extend this by connecting to the internet/network, enabling remote data collection, monitoring, control, and communication with other devices and cloud platforms.
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1b	Significance of Transmission Range and Data Rate of a Sensor Node: Transmission range determines how far a sensor node can send data without signal degradation — critical for coverage area design. Data rate affects how quickly sensor data is transmitted; higher data rates allow more frequent updates but consume more power. Both are key in selecting appropriate wireless technology (e.g., Zigbee, Wi-Fi, LoRa) for an IoT deployment.
1c	WiMAX Link Layer Protocol: IEEE 802.16 is the link layer protocol that is a collection of broadband standards for WiMAX (Worldwide Interoperability for Microwave Access). It supports high-speed wireless broadband access over long distances.
1d	Role of Actuators in IoT: Actuators are output devices that receive control signals from the IoT system and perform physical actions. Examples include motors, solenoid valves, relays, LEDs, and buzzers. They translate digital commands into real-world actions such as opening a valve, turning on a motor, or switching on lights.
1e	Network Layer Protocols in IoT: IPv4, IPv6, 6LoWPAN, RPL (Routing Protocol for Low-Power and Lossy Networks).
1f	Role of IoT in Machine Diagnosis and Prognosis: IoT-enabled sensors collect real-time operational data from machines. This data is used to diagnose faults (current problems) and predict failures before they occur (prognosis). It enables predictive maintenance, reducing downtime and operational costs in manufacturing and industrial environments.

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1a	Transmission Range and Data Rate of a Sensor Node: Transmission range is the maximum distance over which a sensor node can reliably send/receive data. Data rate is the speed of data transmission. These parameters vary by technology: Zigbee (10-100m, 250kbps), LoRa (up to 15km, 0.3-50kbps), Wi-Fi (50-100m, up to 600Mbps). They directly influence power consumption, coverage, and application suitability.
1b (MCQ)	Answer: 3. Transport Layer — The Transport layer provides end-to-end communication in IoT. TCP and UDP are the main transport layer protocols, ensuring reliable or fast data delivery between devices.
1c	Centralized Controller: A centralized controller is a single, central device or system (e.g., a hub, gateway, or server) that manages and coordinates all IoT devices in a network. All devices communicate through this central node, which processes data, applies control logic, and issues commands. It simplifies management but creates a single point of failure.
1d	Protocols used in Transport Layer: TCP (Transmission Control Protocol) — reliable, connection-oriented. UDP (User Datagram Protocol) — fast, connectionless. SCTP (Stream Control Transmission Protocol) — also used in some IoT applications.
1e	Layers in IoT Protocol Suite: The IoT protocol suite typically has 4 layers: Application Layer, Transport Layer, Network Layer, and Data Link/Physical Layer. Some models expand this to 5 or 6 layers including a Middleware or Processing layer.
1f	Drawback of TCP Protocol: TCP is heavy-weight for IoT — it requires a 3-way handshake (connection overhead), has high header size (20 bytes), retransmission mechanisms, and flow control, all of which consume significant bandwidth and energy. This makes TCP unsuitable for constrained, battery-powered IoT devices. UDP or CoAP is preferred instead.

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1a	Embedded System vs IoT: An embedded system is a purpose-built computing system performing specific functions (e.g., microcontroller in a microwave). IoT extends embedded systems with internet connectivity, allowing devices to communicate, share data, be monitored remotely, and integrate with cloud analytics platforms.
1b (MCQ)	Answer: 4. Message Queue Telemetry Transport — MQTT stands for Message Queue Telemetry Transport. It is a lightweight, publish-subscribe protocol ideal for IoT applications with limited bandwidth.
1c	Purpose of NAND/NOR Memory in IoT Devices: NAND flash memory is used for high-density, non-volatile data storage (firmware, logs, large data) in IoT devices. NOR flash is used for code execution in-place (XIP) due to its random-access capability and faster read speeds. NAND is preferred for mass storage; NOR for boot code and critical firmware.
1d	OpenRemote: OpenRemote is an open-source IoT platform for building smart city, smart energy, and building automation applications. It provides tools for device management, data visualization, rule-based automation, and multi-protocol support. It enables integration of heterogeneous IoT devices into a unified management system.
1e (MCQ)	Answer: c) Industrial Applications — Structural health monitoring systems (monitoring bridges, buildings, dams for stress, cracks, vibrations) fall under Industrial Applications / Smart Infrastructure domain of IoT.
1f	Sensor for Humidity and Environment Temperature: DHT11 and DHT22 are popular sensors for combined humidity and temperature measurement. DHT22 offers higher accuracy. BME280 and SHT31 are also used in professional IoT applications for environment monitoring.

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1a	Self-adapting in IoT: Self-adapting (or self-adaptive) refers to the ability of an IoT system to automatically modify its behavior, configuration, or parameters in response to changes in its environment or operating conditions, without human intervention. Example: A smart thermostat that learns user preferences and automatically adjusts temperature schedules based on occupancy patterns and weather data.
1b	Thing in IoT: In IoT, a 'thing' refers to any physical object or device that is embedded with sensors, actuators, and connectivity to collect and exchange data over the internet. It can be anything — a smart bulb, wearable, vehicle, industrial machine, or even a plant. Things are the fundamental building blocks of an IoT ecosystem.
1c	MAC Protocol and Types: MAC (Medium Access Control) protocol manages how devices access and share the communication medium to avoid collisions. Types of MAC protocols in IoT: (1) Schedule-based MAC (TDMA) — assigns fixed time slots (e.g., S-MAC, T-MAC), (2) Contention-based MAC (CSMA) — devices compete for access (e.g., CSMA/CA used in Wi-Fi, Zigbee), (3) Hybrid MAC — combines both approaches for flexibility.

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1a	IoT (Internet of Things): IoT refers to the network of physical devices, vehicles, appliances, and other objects embedded with sensors, software, and connectivity that enables them to collect and exchange data over the internet. It connects the physical
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	world to the digital world, enabling smart monitoring, automation, and data-driven decision-making.
1b	Centralized Controller: A centralized controller is a single controlling entity (like a hub, server, or gateway) through which all IoT devices in a network communicate. It manages device coordination, data processing, and command dispatch. It simplifies control but introduces a single point of failure.
1c	Thing in IoT: A 'thing' in IoT is any physical entity equipped with sensors, actuators, and network connectivity that can sense, communicate, and interact with the environment. Examples: smart meters, connected vehicles, industrial robots, wearables.
1d	IPv4 Address: An IPv4 address is a 32-bit numerical label assigned to each device on a network, written in dotted-decimal format. Example: 192.168.1.10 — it uniquely identifies a device on the internet or local network.
1e	Drawback of TCP Protocol: TCP has significant overhead for IoT — connection setup (3-way handshake), large headers (20 bytes), retransmission, congestion control, and flow control. These features consume too much energy and bandwidth for constrained IoT devices, making UDP or CoAP more suitable alternatives.
1f	Protocols used in Transport Layer: TCP (Transmission Control Protocol) — reliable, ordered delivery with connection setup. UDP (User Datagram Protocol) — fast, connectionless, no delivery guarantee. In IoT, CoAP (Constrained Application Protocol) runs over UDP as a lightweight alternative.

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1a	Embedded System vs IoT: Embedded systems are standalone computing units built for specific tasks without requiring network connectivity. IoT systems extend embedded systems by adding network/internet connectivity, enabling remote monitoring, data aggregation, cloud integration, and inter-device communication for smart applications.
1b	Significance of Transmission Range and Data Rate: Transmission range defines the coverage area of a sensor node — critical for deployment layout. Data rate determines bandwidth and latency of data delivery. Together they guide technology selection: LoRa for long range/low data rate, Wi-Fi for short range/high data rate applications.
1c	WiMAX Link Layer Protocol: IEEE 802.16 is the standard that serves as a broadband wireless communication standard for WiMAX. It provides high-speed internet access over large coverage areas.
1d	Role of Actuators in IoT: Actuators convert digital control signals from the IoT controller into physical actions. They are the output components of an IoT system — examples include servo motors, solenoid valves, relays, pumps, and buzzers. They enable IoT systems to physically interact with and control the real world.
1e	Network Layer Protocols in IoT: IPv4, IPv6, 6LoWPAN (IPv6 over Low-Power Wireless PAN), and RPL (Routing Protocol for Low-Power and Lossy Networks).
1f	Role of IoT in Machine Diagnosis and Prognosis: IoT sensors continuously monitor machine health parameters. Diagnostic analytics identify existing faults, while prognostic analytics predict future failures. This supports predictive and preventive maintenance strategies, improving machine reliability and reducing operational costs.

1a	Role of Controller Service in IoT: The controller service is responsible for receiving data from IoT sensors/devices, processing it based on predefined rules or logic, and issuing control commands to actuators. It acts as the decision-making engine that bridges the sensing and actuation layers of an IoT system.
1b	Things in IoT and Why They Are Important: A 'thing' in IoT is any physical object connected to the internet via sensors/actuators (e.g., smart appliances, vehicles, wearables). Things are important because they are the primary data sources and actuators in an IoT ecosystem — they enable real-world sensing, monitoring, automation, and interaction, making IoT practical and impactful.
1c (MCQ)	Answer: 3. MQTT — MQTT (Message Queue Telemetry Transport) is the lightest protocol among the options. It is designed for constrained IoT devices with minimal bandwidth and power. IP is a network layer protocol; HTTP has high overhead; CoAP is lightweight but MQTT is the most optimized for pub-sub IoT messaging.
1d	Sensors Associated with Smart Watch: Accelerometer (motion/steps), Gyroscope (orientation), Heart Rate Sensor (PPG sensor), GPS (location), Barometer (altitude/pressure), Ambient Light Sensor, SpO2 Sensor (blood oxygen), Temperature Sensor, Microphone.
1e	WBAN (Wireless Body Area Network): WBAN is a wireless network of wearable or implantable sensors and devices placed on or inside the human body. It is used for continuous health monitoring — measuring vital signs like ECG, blood pressure, temperature, and glucose levels. WBANs enable remote patient monitoring and personal health management.
1f	Network Layer Protocols in IoT Protocol Suite: IPv4, IPv6, 6LoWPAN (for low-power devices), RPL (Routing Protocol for Low-Power and Lossy Networks — a distance-vector routing protocol designed for IoT mesh networks).

IoT Mid-Term Exam — Section 2 Complete Answers

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2a — What are the architectural constraints of REST? [3 marks]

REST (Representational State Transfer) defines 6 architectural constraints that a web service must follow:

- 1. Client-Server:** Separation of concerns — the client handles UI, the server handles data/logic. They evolve independently.
- 2. Stateless:** Each client request must contain all information needed. The server stores no session state between requests.
- 3. Cacheable:** Responses must define themselves as cacheable or non-cacheable to improve efficiency and scalability.
- 4. Uniform Interface:** A consistent interface between client and server using standard HTTP methods (GET, POST, PUT, DELETE) and resource URIs.
- 5. Layered System:** Client cannot tell whether it is connected directly to the server or an intermediary (proxy, gateway, load balancer).
- 6. Code on Demand (optional):** Servers can send executable code (e.g., JavaScript) to clients to extend functionality.

2b — Briefly explain the different components of an IoT system. [3 marks]

An IoT system consists of the following key components:

- 1. Sensors/Actuators:** Sensors collect data from the environment (temperature, humidity, motion). Actuators perform physical actions based on control commands (e.g., motor, valve, relay).
- 2. Connectivity/Communication:** Enables data transfer between devices using protocols like Wi-Fi, Zigbee, Bluetooth, LoRa, MQTT, or CoAP.
- 3. IoT Gateway:** Acts as a bridge between IoT devices and the cloud/internet. Handles protocol translation, local data processing, and security.
- 4. Cloud/Backend Platform:** Receives, stores, and processes data. Provides APIs, analytics, and dashboards for data visualization and management.
- 5. Analytics & Processing:** Processes raw data to generate insights, trigger alerts, or feed machine learning models for intelligent decision-making.
- 6. User Interface:** Web/mobile applications that allow users to monitor, control, and interact with IoT devices remotely.

2c — Explain the smart irrigation system with an example. [3 marks]

A smart irrigation system uses IoT to automate watering of crops or gardens based on real-time data, eliminating manual intervention and water wastage.

Components & Working:

- **Sensors:** Soil moisture sensors, temperature sensors, and rain sensors are deployed in the field.
- **Microcontroller/Gateway:** Collects sensor data and sends it to a cloud platform (e.g., via Wi-Fi or LoRa).
- **Decision Logic:** If soil moisture drops below a threshold (e.g., <30%), the system automatically activates the irrigation pump via an actuator (solenoid valve).
- **User Interface:** Farmers can monitor field conditions and override controls remotely through a mobile app.

Example: A paddy farm uses DHT22 + soil moisture sensors. Data is sent to a Raspberry Pi gateway. When moisture < 40%, a relay triggers a water pump. The farmer receives a notification on their smartphone. After irrigation, if moisture > 70%, the pump automatically shuts off — saving ~30% water usage.

2a — What are the functional blocks of IoT? [3 marks]

The IoT functional blocks are the core building components that together enable IoT systems to operate:

- 1. Device (Thing):** Physical objects with sensors and actuators. Collects data and performs actions.
- 2. Communication:** Handles data transfer using protocols like MQTT, CoAP, HTTP, Zigbee, LoRa, Bluetooth.
- 3. Services:** Includes device management, data publishing, subscribing, analytics, and integration services.
- 4. Management:** Manages configuration, monitoring, diagnostics, firmware updates, and security of IoT devices.
- 5. Security:** Ensures authentication, authorization, data encryption, and secure communication across all layers.
- 6. Application:** The end-user layer — dashboards, mobile apps, or enterprise systems that use IoT data for specific use cases.

2b — Mention 3 different technologies with their responsibilities for implementation of IoT. [3 marks]

Three key technologies used in IoT implementation:

- 1. Zigbee:** A low-power, low-data-rate wireless mesh networking protocol (IEEE 802.15.4). Responsible for short-range device-to-device communication in home automation and industrial sensor networks. Range: ~10-100m, Data rate: 250 kbps.
- 2. MQTT (Message Queue Telemetry Transport):** A lightweight publish-subscribe protocol running over TCP/IP. Responsible for reliable, low-overhead messaging between IoT devices and brokers (e.g., cloud servers). Ideal for constrained devices with limited bandwidth.
- 3. LoRa (Long Range):** A long-range, low-power wireless technology. Responsible for transmitting small packets of sensor data over several kilometers (up to 15 km in rural areas). Used in agriculture, smart metering, and city-wide IoT deployments where devices are battery-powered and spread over large areas.

2c — What is the significance of 6LoWPAN protocol? In which layers is it used? [3 marks]

6LoWPAN (IPv6 over Low-Power Wireless Personal Area Networks) is a network protocol that enables IPv6 communication over IEEE 802.15.4 low-power wireless networks.

Significance:

- Allows IPv6 packets (which are large) to be compressed and fragmented to fit the small frame size (127 bytes) of IEEE 802.15.4.
- Enables direct IP connectivity to tiny, battery-powered IoT sensor nodes, eliminating the need for proprietary gateways.
- Supports mesh networking, making it scalable for large sensor deployments like smart cities and industrial IoT.
- Provides header compression, reducing transmission overhead and conserving energy on constrained devices.

Layers Used: 6LoWPAN operates between the Network Layer (IPv6) and the Data Link Layer (IEEE 802.15.4). It acts as an adaptation layer — bridging the gap between these two layers by compressing IPv6 headers and fragmenting large packets.

2a — What are the functional blocks of IoT? [3 marks]

The IoT functional blocks are the essential building units of any IoT system:

- 1. Device:** Physical hardware (sensors + actuators) that interacts with the real world — senses environmental parameters and triggers physical responses.

- 2. Communication:** Protocols and hardware enabling data flow between devices and the cloud (Wi-Fi, Bluetooth, Zigbee, MQTT, CoAP).
- 3. Services:** Middleware services for device discovery, data aggregation, pub-sub messaging, and API management.
- 4. Management:** Device lifecycle management — provisioning, configuration, OTA firmware updates, monitoring, and diagnostics.
- 5. Security:** End-to-end protection through encryption (TLS/SSL), authentication (X.509 certificates), and access control mechanisms.
- 6. Application:** User-facing software (dashboards, mobile apps) that consumes IoT data to deliver business or consumer value.

2b — What is request-response model? [3 marks]

The Request-Response model is a fundamental communication pattern in IoT where a client sends a request to a server, and the server responds with the requested data or acknowledgement.

Working:

- Step 1 — Request:** The client (IoT device or application) initiates communication by sending a request message to the server (e.g., HTTP GET request for sensor data).
- Step 2 — Processing:** The server receives and processes the request, retrieves or computes the required data.
- Step 3 — Response:** The server sends back a response message containing the data or a status code (e.g., 200 OK with JSON payload).

Characteristics:

- Synchronous — the client waits for the server's response before proceeding.
- One-to-one communication pattern.
- Used in REST APIs (HTTP) and CoAP in IoT.
- Simple and widely implemented but less efficient for real-time streaming compared to publish-subscribe.

Example: A mobile app sends an HTTP GET request to an IoT gateway asking for the current temperature reading. The gateway responds with { "temperature": 28.5 }.

2c — Briefly explain the process specifications of the home intrusion detection system. [3 marks]

A Home Intrusion Detection System (HIDS) using IoT detects unauthorized entry and alerts the homeowner in real time.

Process Specifications:

- 1. Sensing:** PIR (Passive Infrared) motion sensors, door/window magnetic reed switches, and glass-break sensors continuously monitor entry points and rooms.
- 2. Data Transmission:** Sensor data is transmitted to a local hub/controller using Zigbee or Wi-Fi. Triggers are sent immediately on detecting an anomaly.
- 3. Controller Processing:** The IoT controller evaluates sensor inputs. If motion is detected while the system is armed, it initiates an alert sequence.
- 4. Alert & Notification:** The system activates a local alarm (buzzer/siren) and simultaneously sends push notifications to the homeowner's smartphone via the cloud.
- 5. Camera Integration:** IP cameras capture live footage which can be streamed to the user and stored in the cloud for evidence.
- 6. Override/Disarm:** The homeowner can disarm the system remotely through a mobile app or using a keypad with PIN/biometric verification.

2a — State various components of an IoT model. [3 marks]

An IoT model consists of the following essential components:

- 1. Physical Devices/Things:** Sensors (collect data) and actuators (perform actions). E.g., temperature sensors, motors.
- 2. Connectivity Layer:** Communication medium — wired (Ethernet) or wireless (Wi-Fi, Zigbee, Bluetooth, LoRa) — for transmitting data.
- 3. IoT Gateway:** Aggregates data from multiple devices, performs local processing, and forwards data to the cloud. Handles protocol conversion.
- 4. Cloud/Data Processing Platform:** Receives data from gateways, stores it, and performs analytics. Examples: AWS IoT, Azure IoT Hub, Google Cloud IoT.
- 5. Analytics Engine:** Processes data using statistical models or machine learning to generate insights, detect anomalies, or trigger automated actions.
- 6. User Interface/Application Layer:** Dashboards, mobile/web apps through which users interact with the IoT system, visualize data, and send control commands.

2b — Summarize the working mechanism of the request-response model. [3 marks]

The Request-Response model is a client-server communication paradigm where all interaction is initiated by the client:

- 1. Client Sends Request:** The client (application or IoT device) sends a request specifying the resource and action (e.g., HTTP GET /sensor/temperature).
- 2. Server Receives & Processes:** The server identifies the resource, performs the required operation (read/write data), and prepares the response.
- 3. Server Sends Response:** The server replies with the result — typically a status code and data payload (e.g., 200 OK + JSON data).
- 4. Connection Closed:** In HTTP/1.0, the connection is closed after each exchange. HTTP/1.1 supports keep-alive for efficiency.

Key Features: Synchronous, stateless (in REST), scalable. Used in CoAP and HTTP-based IoT APIs. Suitable for on-demand data retrieval rather than continuous real-time streaming.

2c — Explain the service specifications of the state service for home automation IoT systems with an appropriate diagram. [3 marks]

The State Service in a home automation IoT system maintains the current status (state) of all connected devices and provides mechanisms to read and update these states.

Service Specifications:

State Representation: Each device (bulb, thermostat, lock) has a state object — e.g., { 'bulb_1': 'ON', 'temp': 22, 'lock': 'LOCKED' }.

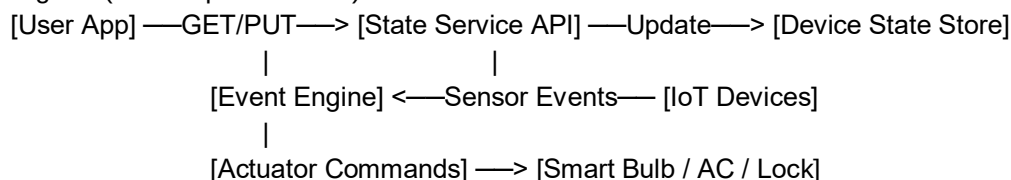
Read State (GET): Client apps query the state service to retrieve the current state of a device or the entire home system.

Write/Update State (PUT/POST): Users or automation rules update device states. E.g., setting 'bulb_1' to 'OFF' triggers the actuator command.

Event-Driven Updates: When physical changes occur (e.g., motion detected), the state service is updated in real time, and subscribed clients are notified.

Persistence: States are stored persistently so the system can restore last known states after a reboot or power failure.

Diagram (Text Representation):



2a — What are the components of an IoT model? [3 marks]

The major components of an IoT model are:

- 1. Sensors & Actuators:** Sensors gather environmental data (light, temperature, pressure). Actuators respond to commands (motors, valves, relays).
- 2. Communication Network:** Wireless or wired protocols (Zigbee, LoRa, Wi-Fi, MQTT, CoAP) that carry data between devices and the cloud.
- 3. IoT Gateway/Hub:** Bridges devices to the internet. Performs data aggregation, filtering, and protocol translation.
- 4. Cloud Platform:** Stores data, performs analytics, and exposes APIs. Examples: AWS IoT Core, Azure IoT Hub.
- 5. Data Analytics:** Transforms raw data into actionable insights using statistical analysis, ML models, or rule engines.
- 6. Application Layer:** End-user interfaces (mobile apps, web dashboards) for monitoring, control, and reporting.

2b — Differentiate between sensor and actuator. [3 marks]

Sensor vs Actuator:

Parameter	Sensor	Actuator
Definition	Detects physical/environmental changes and converts them into electrical signals	Converts electrical signals/commands into physical actions
Function	Input device — collects data from environment	Output device — performs actions in the environment
Direction	Environment → System (data in)	System → Environment (action out)
Examples	DHT22 (temp), PIR (motion), LDR (light), MQ-2 (gas)	Servo motor, relay, solenoid valve, LED, buzzer
IoT Role	Provides data for monitoring and analytics	Enables automation and control

2c — What is the significance of LOWPAN protocol? In which layers is it used? [3 marks]

6LoWPAN (IPv6 over Low-Power Wireless Personal Area Networks) enables IPv6 communication on IEEE 802.15.4 low-power, short-range wireless networks.

Significance:

- Compresses large IPv6 headers (40 bytes) to fit within the 127-byte IEEE 802.15.4 frame size, enabling IP connectivity on tiny IoT nodes.
- Enables direct internet addressability for each sensor node using IPv6 — no need for custom gateways.
- Supports mesh networking and multi-hop routing, extending the coverage of WSNs.
- Reduces power consumption by minimizing packet overhead, extending battery life of sensor nodes.

Layer Used: 6LoWPAN is an adaptation layer that sits between the Network Layer (IPv6) and the Data Link Layer (IEEE 802.15.4). It serves as a glue layer enabling the two incompatible layers to work together.

2d — Why is network wide configuration important for IoT systems with multiple nodes? [3 marks]

Network-wide configuration ensures all nodes in an IoT system operate consistently, securely, and efficiently as a coordinated unit:

- 1. Consistent Communication:** All nodes must share the same network parameters (IP addresses, channel, PAN ID, security keys) to communicate reliably. Misconfiguration causes connectivity failures.

- 2. IP Address Management:** With potentially thousands of devices, centralized configuration (e.g., via DHCP or SLAAC with 6LoWPAN) prevents address conflicts and simplifies tracking.
- 3. Security Enforcement:** Network-wide policies ensure all nodes use the same encryption standards and authentication mechanisms, preventing unauthorized devices from joining.
- 4. Scalability:** Centralized configuration tools (e.g., network management servers) allow administrators to onboard, update, or retire nodes at scale without manual per-device setup.
- 5. Fault Isolation:** Uniform configuration makes it easier to identify and isolate misbehaving nodes, simplifying troubleshooting in large deployments.

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2a — What are the architectural constraints of REST? [3 marks]

REST (Representational State Transfer) is governed by 6 architectural constraints:

- 1. Client-Server:** UI (client) and data/logic (server) are separated. This allows them to evolve independently, improving portability and scalability.
- 2. Stateless:** No session is stored on the server. Each request from the client must contain all context needed to process it. Improves reliability and scalability.
- 3. Cacheable:** Server responses must declare if they are cacheable. Caching reduces server load and improves performance.
- 4. Uniform Interface:** Standardized interaction via resource URIs, HTTP methods (GET/POST/PUT/DELETE), and self-descriptive messages. Simplifies and decouples architecture.
- 5. Layered System:** The architecture can include intermediary layers (load balancers, caches, gateways) that are transparent to the client.
- 6. Code on Demand (optional):** Servers may send executable scripts (like JavaScript) to clients to extend functionality dynamically.

2b — What are the components of an IoT model? [3 marks]

The components of an IoT model are:

- 1. Things (Devices):** Physical objects embedded with sensors and actuators. The primary data collection and action point of the system.
- 2. Communication Network:** Wired/wireless medium for data transport — Wi-Fi, Ethernet, LoRa, Zigbee, Bluetooth, Cellular.
- 3. Gateway:** Aggregates data from devices, performs local pre-processing, and provides internet connectivity.
- 4. Cloud Platform:** Centralized backend for data storage, processing, and API services — e.g., AWS IoT, Google Cloud IoT.
- 5. Analytics & Intelligence:** Extracts meaningful information from data using analytics, ML, and AI techniques.
- 6. Application:** User-facing interfaces for monitoring, alerts, and control (web apps, mobile apps, dashboards).

2c — Explain the service specifications of the mode service for the home automation IoT system with an appropriate diagram. [3 marks]

The Mode Service in a home automation system allows users to activate predefined operational modes that configure multiple devices simultaneously.

Common Modes:

- **Home Mode:** Normal operation — lights on, AC at comfort setting, security disarmed.
- **Away Mode:** Lights off, AC off, security system armed, energy saving enabled.
- **Sleep Mode:** Dim lights, lower AC temperature, lock doors, silence notifications.
- **Party Mode:** Bright lights, entertainment system on, relaxed security settings.

Service Specifications:

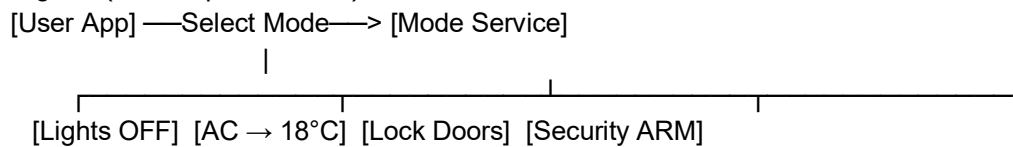
Mode Activation: User selects a mode via mobile app → Mode Service receives the command.

Device Mapping: Mode Service maps the selected mode to specific device states (e.g., Away Mode → all lights OFF, thermostat to 18°C).

State Update: Mode Service sends commands to individual device controllers/actuators.

Confirmation: Mode Service collects acknowledgements from devices and confirms activation to user.

Diagram (Text Representation):



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2a — What are the functional blocks of IoT? [3 marks]

IoT functional blocks describe the logical functions that any IoT system must implement:

- 1. Device (Thing):** Sensors and actuators that interact with the physical world — sense data and perform actions.
- 2. Communication:** Enables data transfer using wireless/wired protocols (MQTT, Zigbee, Wi-Fi, LoRa, Bluetooth, CoAP).
- 3. Services:** Software services for device discovery, data publishing/subscribing, identity management, and system integration.
- 4. Management:** Handles provisioning, monitoring, diagnostics, configuration, and OTA (Over-the-Air) updates.
- 5. Security:** Provides authentication, encryption, authorization, and intrusion detection to protect data and devices.
- 6. Application:** The end-user layer that consumes IoT data for specific applications — smart home, healthcare, agriculture, etc.

2b — Differentiate between TCP and UDP. [3 marks]

TCP vs UDP comparison:

Parameter	TCP	UDP
Full Form	Transmission Control Protocol	User Datagram Protocol
Connection	Connection-oriented (3-way handshake)	Connectionless — no setup required
Reliability	Guaranteed delivery with retransmission	No delivery guarantee — best effort
Ordering	Data delivered in order	No ordering guarantee
Speed	Slower due to acknowledgements	Faster — minimal overhead
Header Size	20 bytes (heavy)	8 bytes (lightweight)
IoT Use Cases	REST API, MQTT (reliable messaging)	CoAP, video streaming, DNS
Error Checking	Yes — with correction	Yes — detection only, no correction

2c — What is IoT level 1? [3 marks]

IoT Deployment Level 1 is the simplest IoT architecture involving a single device that performs sensing, analysis, and actuation — all locally without a gateway or cloud.

Characteristics:

- **Single Node:** One device with embedded sensors, local processing, and actuation capability.
- **No Cloud/Internet:** Data is processed locally on the device — no external connectivity required.
- **Local Storage:** Data may be stored locally (on device memory or SD card).
- **Direct Visualization:** Output is displayed locally — e.g., on an LCD screen attached to the device.
- **Low Complexity:** Minimal infrastructure, easy to deploy, suitable for simple standalone applications.

Example: A standalone digital thermometer/hygrometer with a display that reads temperature and humidity, stores readings locally, and triggers a buzzer if temperature exceeds a threshold — no internet connection needed.

Diagram: [Sensor] → [Microcontroller/Processing] → [Local Display/Actuator]

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2a — Differentiate wired and wireless medium for communication. [3 marks]

Wired vs Wireless Communication in IoT:

Parameter	Wired Communication	Wireless Communication
Medium	Physical cables — Ethernet, coaxial, fiber optic	Air/radio waves — Wi-Fi, Zigbee, LoRa, Bluetooth
Speed	High (Gbps range for Ethernet)	Lower than wired; varies by technology
Reliability	High — less interference	Moderate — subject to noise, interference
Mobility	Fixed — limited mobility	High mobility — devices can move freely
Cost	Higher installation cost (cabling)	Lower deployment cost, especially over large areas
Security	More secure (physical access needed)	More vulnerable to eavesdropping/jamming
IoT Suitability	Industrial IoT, server rooms, fixed sensors	Smart homes, wearables, agriculture, smart cities
Examples	Modbus, RS-485, Ethernet, CAN bus	Wi-Fi (802.11), Zigbee, LoRa, Bluetooth, Cellular

2b — What is IoT level 1? [3 marks]

IoT Level 1 is the most basic deployment model where a single device with embedded sensing, processing, and actuation operates completely locally without cloud or internet connectivity.

Key Features:

- Single device performs all functions: sensing → analysis → action.
- No gateway, no cloud, no internet required.
- Data is processed and stored locally (on-device or local storage).
- Output displayed or acted upon locally (LED, LCD, buzzer).
- Low power, low cost, minimal complexity.

Suitable For: Simple monitoring and control systems where remote access is not needed.

Example: A soil moisture sensor connected to an Arduino that automatically triggers a water pump when moisture is low — all processing happens on the Arduino with no internet connection.

Architecture: [Sensor] → [Local Processor (Arduino/Pi)] → [Actuator + Local Display]

2c — Why is network wide configuration important for IoT systems with multiple nodes? [3 marks]

Network-wide configuration is critical for managing, securing, and scaling IoT deployments with many nodes:

- 1. Uniform Communication Parameters:** All nodes must share the same network ID, channel, and addressing scheme to communicate without conflicts.
- 2. IP Address Management:** Prevents address collisions and enables efficient routing. DHCP or IPv6 SLAAC with 6LoWPAN handles this at scale.
- 3. Security Consistency:** Ensures all nodes enforce the same authentication, encryption, and access control policies — preventing rogue devices.
- 4. Centralized Firmware/Config Updates:** OTA (Over-the-Air) updates can be pushed to all nodes simultaneously, reducing manual effort.
- 5. Monitoring & Diagnostics:** A uniformly configured network allows central monitoring systems to track all nodes, detect failures, and generate alerts.
- 6. Scalability:** As the number of nodes grows, centralized configuration management prevents chaos and ensures the system remains maintainable.

2d — What are the various network layer protocols associated in IoT? [3 marks]

The Network Layer in IoT is responsible for logical addressing and routing of data packets. Key protocols include:

- 1. IPv4:** 32-bit addressing protocol. Widely used but limited address space (4.3 billion addresses) — insufficient for the massive scale of IoT.
- 2. IPv6:** 128-bit addressing providing virtually unlimited addresses (3.4×10^{38}). Essential for the billions of IoT devices needing unique global addresses.
- 3. 6LoWPAN:** Enables IPv6 over IEEE 802.15.4 low-power networks by compressing IPv6 headers and fragmenting packets. Acts as an adaptation layer for constrained devices.
- 4. RPL (Routing Protocol for Low-Power and Lossy Networks):** A distance-vector IPv6 routing protocol designed for IoT mesh networks with unreliable links and constrained nodes. Builds a DODAG (Destination-Oriented Directed Acyclic Graph) for routing.
- 5. ROLL:** Routing Over Low-power and Lossy networks — framework that defines requirements for routing in IoT environments.

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2a — State various components of an IoT model. [3 marks]

The key components of an IoT model are:

- 1. Sensing Layer (Devices):** Sensors collect physical data (temperature, pressure, motion). Actuators execute control commands.
- 2. Network/Communication Layer:** Transmits data between devices and the backend using Wi-Fi, Zigbee, LoRa, Bluetooth, Ethernet, or cellular networks.
- 3. Gateway:** Edge device that aggregates data, performs local pre-processing, and provides secure internet connectivity.
- 4. Cloud/Data Platform:** Centralized server for storing, processing, and analyzing data from all devices.
- 5. Analytics & AI Engine:** Processes data for insights, predictions, anomaly detection, and automated decision-making.
- 6. User Application:** Dashboard/mobile app for visualization, alerts, reports, and device control.

2b — Summarize the working mechanism of the request-response model. [3 marks]

The Request-Response model is a synchronous, client-server communication pattern used in IoT APIs (HTTP, CoAP):

- Step 1 — Initiation:** The client (e.g., mobile app or IoT controller) sends a request to the server with a method (GET/POST/PUT) and target resource URI.

Step 2 — Server Processing: The server receives the request, authenticates the client, processes the request (queries database or controls a device), and prepares the response.

Step 3 — Response Delivery: The server sends back a response — including a status code (200 OK, 404 Not Found, etc.) and a data payload (JSON/XML).

Step 4 — Client Action: The client processes the response — displays data to the user or triggers further actions.

Properties: Stateless, synchronous, scalable. Ideal for on-demand data retrieval. Example: HTTP GET /api/temperature → Response: {"value": 24.5, "unit": "C"}

2c — Explain the service specifications of the state service for home automation IoT systems with an appropriate diagram. [3 marks]

The State Service maintains and manages the real-time state of all devices in a home automation system.

Core Specifications:

State Storage: A persistent data store holds the current state of every device — e.g., { 'fan': 'OFF', 'ac_temp': 22, 'door_lock': 'LOCKED' }.

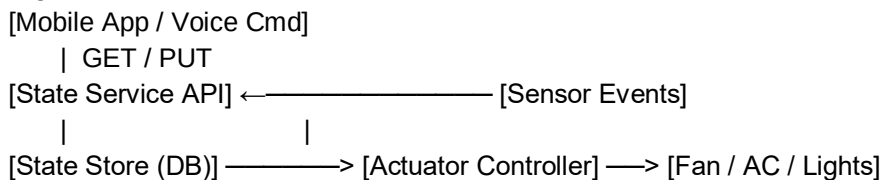
State Read (GET): Client applications query the state service to check current device status before displaying or deciding next actions.

State Write (PUT): User commands or automation rules update states. Writing 'fan: ON' triggers the fan actuator.

Event-Driven Sync: Hardware state changes (e.g., manual switch press) are reflected in the state store immediately via event listeners.

State History: Previous states are logged for energy auditing and automation learning.

Diagram:



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2a — Differentiate between TCP and UDP. [3 marks]

TCP (Transmission Control Protocol) vs UDP (User Datagram Protocol):

Parameter	TCP	UDP
Connection	Connection-oriented (3-way handshake)	Connectionless
Reliability	Reliable — guarantees delivery & order	Unreliable — best effort, no guarantee
Speed	Slower (acknowledgements, retransmission)	Faster (no overhead)
Header	20 bytes	8 bytes
Flow Control	Yes (congestion & flow control)	No
IoT Use	MQTT, HTTP REST APIs — reliable data	CoAP, live streaming — speed-critical
Error Correction	Yes — automatic retransmission	Detection only, no correction

For IoT: UDP/CoAP is preferred on constrained devices to save energy; TCP is used where data integrity is critical (e.g., MQTT over TCP for telemetry).

2b — What are the functional blocks of IoT? [3 marks]

The functional blocks describe all the capabilities that must be present in a complete IoT system:

- 1. Device:** The physical 'thing' with sensors (input) and actuators (output). It is the interface between the digital and physical worlds.
- 2. Communication:** Handles data transport using appropriate protocols — MQTT, CoAP, Zigbee, LoRa, Wi-Fi, Bluetooth — based on range, power, and bandwidth needs.
- 3. Services:** Cloud or edge services for device management, data ingestion, messaging (pub-sub), and API exposure.
- 4. Management:** Device provisioning, configuration management, remote diagnostics, OTA firmware updates, and lifecycle management.
- 5. Security:** Identity verification (authentication), access control (authorization), data encryption (TLS/SSL), and attack prevention.
- 6. Application:** Business logic and user interfaces — dashboards, mobile apps, and enterprise integrations that deliver value from IoT data.

2c — What is the significance of 6LoWPAN protocol? In which layers is it used? [3 marks]

6LoWPAN (IPv6 over Low-Power Wireless Personal Area Networks) is a critical adaptation protocol that makes IPv6 connectivity possible on resource-constrained IoT devices.

Significance:

- IPv6 packets are 40 bytes in header size alone, far too large for the 127-byte IEEE 802.15.4 frame. 6LoWPAN compresses IPv6 headers to as few as 2 bytes.
- Enables every IoT sensor node to have a unique global IPv6 address — crucial as the number of devices exceeds IPv4 capacity.
- Supports packet fragmentation and reassembly for payloads that exceed the maximum frame size.
- Enables mesh networking and multi-hop communication for wider coverage without additional gateways.
- Reduces transmission overhead → saves battery power → extends sensor node lifetime in the field.

Layers: 6LoWPAN operates as an Adaptation Layer between the Network Layer (IPv6) and the Data Link Layer (IEEE 802.15.4 MAC). It acts as a translator/compressor enabling IPv6 to work seamlessly over low-power wireless hardware.

Stack Position: [Application] → [Transport (UDP)] → [Network (IPv6)] → [6LoWPAN Adaptation] → [Data Link (IEEE 802.15.4)] → [Physical]

IoT Mid-Term Exam — Sections 3 & 4 Complete Answers (Pages 1–5)

Silicon University | BTCS-T-PC-056 | 5 Marks Each Question

Page 1 — 7th Sem B.Tech Mid-Term 2025-26 (Repeat)

SECTION 3 — Answer Any One

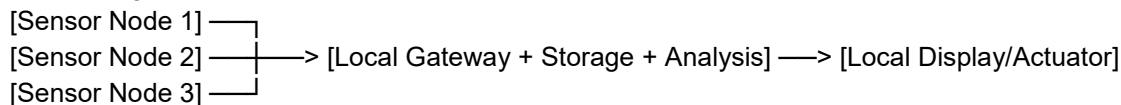
3a — Differentiate the IoT Deployment Level-3 and Level-4 with diagram. [5 marks]

IoT deployment levels define the architectural complexity of IoT systems. Level-3 and Level-4 differ primarily in where data analysis occurs and how many nodes are involved.

IoT Level-3:

- **Architecture:** Multiple end nodes → Local gateway (with storage & analysis) → No cloud required.
- **Analysis:** Data analysis happens locally at the gateway or edge device.
- **Storage:** Data is stored locally on the gateway (e.g., SD card, local database).
- **Connectivity:** Devices communicate via short-range protocols (Zigbee, Bluetooth, Wi-Fi) to a local gateway.
- **Example:** A factory floor with multiple temperature sensors → Local industrial PC analyzes data → Triggers alerts locally.

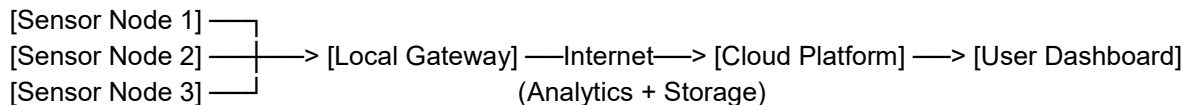
Level-3 Diagram:



IoT Level-4:

- **Architecture:** Multiple end nodes → Local gateway → Cloud server for analysis.
- **Analysis:** Data is forwarded to a cloud platform for deeper analysis, ML processing, and long-term storage.
- **Storage:** Cloud-based storage (e.g., AWS S3, Azure Blob Storage).
- **Connectivity:** Gateway connects to the internet; devices use short-range protocols to gateway.
- **Difference from Level-3:** Level-4 adds cloud connectivity and remote access; Level-3 is entirely local.
- **Example:** Smart weather station → Raspberry Pi gateway → AWS IoT → Dashboard accessible globally.

Level-4 Diagram:



Feature	Level-3	Level-4
Analysis Location	Local gateway/edge	Cloud server
Internet Required	No	Yes
Remote Access	Not available	Available from anywhere
Storage	Local (limited)	Cloud (scalable)
Complexity	Medium	Higher
Example	Factory floor monitor	Smart weather station with cloud dashboard

3b — Discuss the various communication models in IoT. [5 marks]

IoT communication models define how devices exchange data. There are four primary models:

1. Request-Response Model

- **Description:** Client sends a request; server processes and returns a response. Synchronous communication.
- **Protocol:** HTTP, CoAP.
- **Use Case:** Fetching sensor readings on demand. E.g., mobile app requests current temperature from a weather node.
- **Diagram:** [Client] —Request—> [Server] —Response—> [Client]

2. Publish-Subscribe Model

- **Description:** Publishers send messages to a broker on specific topics. Subscribers register interest in topics and receive messages when published. Decoupled — publisher and subscriber don't know each other.
- **Protocol:** MQTT, AMQP.
- **Use Case:** Real-time sensor telemetry. E.g., temperature sensor publishes to topic 'home/temp'; app subscribes and receives data instantly.
- **Diagram:** [Publisher] —topic—> [Broker] —topic—> [Subscriber 1, 2, N]

3. Push-Pull Model

- **Description:** Producers push data into a queue; consumers pull data from the queue at their own pace. Asynchronous and decoupled.
- **Protocol:** ZeroMQ, AMQP.
- **Use Case:** Load balancing across multiple IoT data consumers. E.g., a pipeline where sensor data is queued and multiple processors pull tasks.
- **Diagram:** [Producer] —push—> [Queue] <—pull— [Consumer 1, 2, N]

4. Exclusive Pair Model

- **Description:** A persistent, full-duplex, bidirectional connection between a client and server. Both can send messages at any time.
- **Protocol:** WebSocket.
- **Use Case:** Real-time interactive control. E.g., browser-based dashboard receiving live sensor updates without polling.
- **Diagram:** [Client] <====bidirectional====> [Server]

SECTION 4 — Answer Any One

4a — Describe the use of IoT in smart cities. [5 marks]

A smart city uses IoT to improve the quality of life for citizens, optimize resource usage, and enhance urban services through real-time data collection and automation.

1. Smart Traffic Management

- **Sensors & Cameras:** Detect vehicle density at intersections. Traffic signals adjust timing dynamically to reduce congestion.
- **Example:** Adaptive traffic light systems using vehicle counting sensors reduce average wait time by 25%.

2. Smart Street Lighting

- **IoT Lights:** Street lamps equipped with light sensors and motion detectors dim or brighten based on ambient light and pedestrian presence.
- **Benefit:** Reduces energy consumption by up to 50%.

3. Smart Waste Management

- **Smart Bins:** Ultrasonic sensors detect fill levels and report to a central system. Collection trucks are dispatched only when needed.
- **Benefit:** Reduces unnecessary trips, saving fuel and reducing emissions.

4. Smart Water Management

- **Water Meters:** IoT-enabled meters monitor consumption, detect leaks, and report anomalies in real time.
- **Pipeline Sensors:** Pressure and flow sensors detect pipe bursts instantly.

5. Environmental Monitoring

- **Air Quality Sensors:** Monitor PM2.5, CO2, NO2 levels across the city. Data fed to dashboards and apps to warn citizens of pollution.

6. Smart Parking

- **Parking Sensors:** Detect occupied/vacant spots. Mobile apps guide drivers to available spots, reducing circling traffic.

7. Public Safety & Surveillance

- **Cameras & Gunshot Detectors:** AI-powered cameras and acoustic sensors detect incidents and alert authorities automatically.

Overall Impact: IoT in smart cities reduces operational costs, lowers energy and resource consumption, enhances citizen safety, and enables data-driven governance.

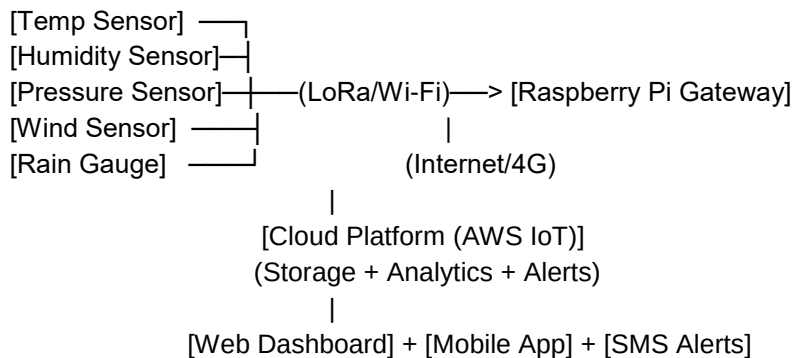
4b — Explain the IoT deployment level required to implement a weather monitoring system with a neat diagram. [5 marks]

A weather monitoring system requires IoT Deployment Level-4 — multiple sensor nodes feeding data through a gateway to a cloud platform for real-time analysis and global accessibility.

Components & Architecture:

- **Sensing Layer (Nodes):** DHT22 (temperature & humidity), BMP280 (atmospheric pressure), anemometer (wind speed), rain gauge, UV sensor, and pyranometer (solar radiation).
- **Communication:** Sensor nodes transmit data using LoRa (long range) or Wi-Fi to a Raspberry Pi / Arduino gateway.
- **Gateway (Level-3 element):** Local pre-processing, data formatting (JSON), and internet uplink via Ethernet or 4G/LTE modem.
- **Cloud Platform (Level-4):** Data sent to AWS IoT / ThingSpeak. Cloud stores data, performs trend analysis, generates forecasts using ML models.
- **User Interface:** Web dashboard and mobile app display real-time readings, historical charts, weather alerts, and forecasts.

Level-4 Deployment Diagram:



Why Level-4?

- Multiple geographically distributed sensor nodes require a gateway for aggregation.
- Cloud analysis enables weather prediction, historical trend analysis, and data sharing via public APIs.
- Remote accessibility is essential for meteorological stations.

SECTION 3 — Answer Any One

3a — Differentiate between REST and WebSocket APIs. [5 marks]

REST and WebSocket are both web-based communication approaches used in IoT, but they serve different purposes and have distinct characteristics.

Parameter	REST API	WebSocket API
Communication Model	Request-Response (client initiates)	Full-duplex bidirectional (both sides can initiate)
Connection	Stateless — new connection per request (HTTP/1.0) or persistent with keep-alive (HTTP/1.1)	Persistent, stateful connection maintained throughout session
Protocol	HTTP/HTTPS	WebSocket (ws:// or wss://) — upgrades from HTTP
Data Flow	One direction per request — client pulls data	Both client and server push data freely at any time
Latency	Higher — repeated connection overhead	Lower — connection established once, messages exchanged instantly
Overhead	Heavy HTTP headers on every request	Minimal overhead after initial handshake
Scalability	Highly scalable due to statelessness	Harder to scale — server must maintain open connections
IoT Use Case	On-demand data retrieval, device config APIs	Real-time sensor streaming, live dashboards, control applications
Example Protocol	CoAP, HTTP GET /temperature	WebSocket streaming of live ECG data

When to Use in IoT:

- **REST:** When data is requested periodically (e.g., polling a smart meter every 5 minutes). Simpler to implement and debug.
- **WebSocket:** When real-time, bidirectional communication is needed (e.g., live industrial monitoring dashboard, interactive robot control).

3b — Describe all IoT communication models with diagrams. [5 marks]

(See Page 1 Section 3b for full description of all 4 models: Request-Response, Publish-Subscribe, Push-Pull, Exclusive Pair — answers are identical in scope.)

1. Request-Response

- **How it works:** Client sends a request; server responds. Synchronous, one-to-one.
- **Protocol:** HTTP, CoAP. Use case: on-demand sensor data retrieval.
- [Client] —GET /sensor—> [Server] —{"temp":24}—> [Client]

2. Publish-Subscribe

- **How it works:** Publishers push messages to a broker on named topics. Subscribers receive messages asynchronously. Decoupled.
- **Protocol:** MQTT, AMQP. Use case: real-time telemetry, smart home automation.
- [Sensor] —publish 'home/temp'—> [MQTT Broker] —> [App 1] [App 2]

3. Push-Pull

- **How it works:** Producers push data into a pipeline/queue. Consumers pull when ready. Load-balanced.
- **Protocol:** ZeroMQ. Use case: distributed data processing pipelines.
- [Sensor Data Producer] —> [Queue/Pipeline] <— [Worker 1] [Worker 2]

4. Exclusive Pair (WebSocket)

- **How it works:** Permanent bidirectional connection between exactly one client and one server. Real-time, low latency.

- **Protocol:** WebSocket. Use case: live dashboards, robot control, video streaming.
- [Browser Client] <== full-duplex WebSocket ==> [IoT Server]

SECTION 4 — Answer Any One

4a — Explain the various components of smart farming and their uses. [5 marks]

Smart farming integrates IoT sensors, automation, data analytics, and connectivity to optimize agricultural productivity and resource usage.

Key Components:

- **1. Soil Monitoring Sensors:** Measure soil moisture, pH, temperature, and nutrient levels. Data helps optimize irrigation schedules and fertilizer application. Examples: capacitive soil moisture sensors, EC (electrical conductivity) sensors.
- **2. Weather Stations:** Collect localized meteorological data — temperature, humidity, wind speed, rainfall, solar radiation. Used to predict irrigation needs and pest activity.
- **3. Irrigation Control System:** Automated drip or sprinkler systems activated based on soil moisture data. Solenoid valves controlled by microcontrollers. Reduces water waste by 30–40%.
- **4. Crop Health Monitoring (Drones + Cameras):** Multispectral cameras on drones analyze NDVI (Normalized Difference Vegetation Index) to detect crop stress, disease, or pest infestations early.
- **5. Livestock Monitoring:** RFID tags and GPS collars track animal location, health metrics (body temperature, activity), and feeding patterns.
- **6. IoT Gateway/Hub:** Aggregates data from all field sensors and communicates with the cloud over LoRa or cellular networks.
- **7. Cloud Platform & Analytics:** Stores farm data, provides dashboards, runs ML models for yield prediction, disease forecasting, and resource optimization.
- **8. Mobile Application:** Allows farmers to monitor field conditions, receive alerts, and control irrigation remotely from a smartphone.

Benefits: Reduced input costs, increased yield, water conservation, early disease detection, and data-driven precision agriculture.

4b — Describe the Application Layer in the IoT protocol suite. [5 marks]

The Application Layer is the topmost layer of the IoT protocol suite. It defines the protocols and interfaces through which IoT applications exchange data with devices and each other.

Role of Application Layer:

- Provides communication between IoT devices and end-user applications.
- Defines message format, encoding (JSON, XML, CBOR), and semantics.
- Handles service discovery, data representation, and application-level security.

Key Application Layer Protocols in IoT:

- **1. HTTP/HTTPS:** Standard web protocol. Used for RESTful APIs. Heavy for constrained devices but widely supported. Port 80/443.
- **2. MQTT (Message Queue Telemetry Transport):** Lightweight publish-subscribe protocol over TCP. Ideal for low-bandwidth, high-latency IoT networks. Uses a broker to decouple producers and consumers. Port 1883/8883.
- **3. CoAP (Constrained Application Protocol):** Lightweight REST-like protocol designed for constrained IoT devices. Runs over UDP. Supports observe mode (server pushes updates to client). Port 5683.
- **4. AMQP (Advanced Message Queuing Protocol):** Enterprise-grade messaging protocol with queuing, routing, and reliability features. Used in industrial IoT.
- **5. WebSocket:** Full-duplex protocol for real-time bidirectional communication. Used in live dashboards and interactive IoT applications.
- **6. XMPP (Extensible Messaging and Presence Protocol):** Used for real-time messaging and presence information in IoT systems.
- **7. DDS (Data Distribution Service):** Pub-sub protocol for real-time, scalable, high-performance IoT systems like autonomous vehicles and robotics.

Comparison:

Protocol	Transport	Pattern	Best For
MQTT	TCP	Pub-Sub	Low-power sensors, telemetry
CoAP	UDP	Req-Resp	Constrained devices, M2M
HTTP	TCP	Req-Resp	Web APIs, non-constrained devices
WebSocket	TCP	Exclusive Pair	Real-time dashboards
AMQP	TCP	Pub-Sub/Queue	Enterprise IoT, reliable delivery

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SECTION 3 — Answer Any One

3a — Differentiate between REST and WebSocket APIs. [5 marks]

(Full comparison same as Page 2 Section 3a — see above for detailed table.)

Key Differences Summary:

- **REST:** Stateless, request-response, HTTP-based, client initiates every exchange, higher latency, easier to scale, suitable for periodic/on-demand IoT queries.
- **WebSocket:** Stateful, full-duplex, persistent connection, either side can send at any time, very low latency, harder to scale, ideal for real-time IoT streaming, live monitoring, and interactive control.

Additional Technical Details:

- **REST Statelessness:** Each request is independent. No server-side session. Excellent for RESTful IoT device management APIs (create device, update firmware, read telemetry).
- **WebSocket Handshake:** Begins with an HTTP Upgrade request. Once established, the connection switches to the WebSocket protocol, reducing per-message overhead to just 2–10 bytes framing overhead vs. 200+ bytes for HTTP headers.
- **IoT Example — REST:** Smart home app polls `/api/livingroom/lights` every 10 seconds to check status.
- **IoT Example — WebSocket:** Industrial monitor maintains a WebSocket connection to a PLC and receives vibration data 100 times/second in real time.

3b — Describe the IoT Deployment Level-1 and Level-2 with clear diagram. Illustrate by taking an IoT application in each case. [5 marks]

IoT Deployment Level-1:

- **Definition:** The simplest IoT architecture — a single node with sensors, local processing, and actuation. No internet or gateway required.
- **Characteristics:** Single device; local data analysis; local storage (SD/EEPROM); output on local display or buzzer; no remote access.
- **Application Example:** Standalone digital thermometer — DHT22 sensor connected to Arduino, displays temperature/humidity on LCD, triggers a buzzer if temperature exceeds 40°C. No internet needed.

Level-1 Diagram:

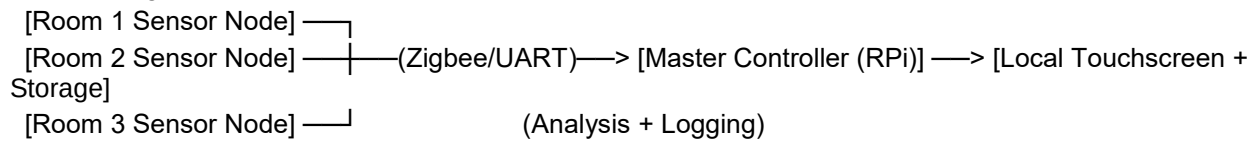
[DHT22 Sensor] —> [Arduino Uno (Processing)] —> [LCD Display + Buzzer]
(Sense) (Analyze & Decide) (Actuate/Display)

IoT Deployment Level-2:

- **Definition:** Multiple sensor nodes connected to a single local controller/master node with local analysis and storage. Still no cloud/internet, but involves multiple nodes.

- **Characteristics:** Multiple sensor nodes; one central controller (master); local network (Zigbee, I2C, SPI, UART); local data storage and display; no remote access.
- **Application Example:** Multi-room temperature monitoring system — DHT22 sensors in 3 rooms each connected to Arduino slaves; a Raspberry Pi master collects all readings, stores them in a local CSV file, and displays on a touchscreen.

Level-2 Diagram:



Feature	Level-1	Level-2
Number of Nodes	Single node	Multiple nodes + master
Network	None	Local (Zigbee/SPI/UART)
Analysis	On-device	Central local controller
Internet	No	No
Example	Standalone thermometer	Multi-room monitor

SECTION 4 — Answer Any One

4a — Describe the working procedure of retail management using IoT. [5 marks]

IoT transforms retail management by automating inventory tracking, enhancing customer experience, optimizing energy use, and providing real-time business intelligence.

1. Smart Inventory Management

- **RFID Tags:** Every product has an RFID tag. RFID readers at shelves and stockrooms automatically track item quantities in real time.
- **Auto-Reordering:** When stock falls below a threshold, the system automatically places an order with suppliers via ERP integration.
- **Smart Shelves:** Weight sensors on shelves detect when items are removed and update inventory instantly.

2. Customer Analytics

- **Footfall Counters:** IR sensors or cameras at entry/exit points count customers and analyze peak hours.
- **Heat Mapping:** Bluetooth beacons or cameras track customer movement patterns to identify high-traffic zones for product placement optimization.

3. Smart Checkout

- **RFID Self-Checkout:** Customers place items in a checkout zone; RFID readers scan all items simultaneously for fast, frictionless checkout.
- **Loss Prevention:** Smart cameras with AI detect shoplifting behavior and alert security staff.

4. Energy Management

- **Smart HVAC & Lighting:** Occupancy sensors adjust air conditioning and lighting based on store activity levels, reducing energy costs by up to 30%.

5. Cold Chain Monitoring

- **Refrigeration Sensors:** Temperature/humidity sensors in cold storage areas monitor conditions 24/7 and alert staff if thresholds are breached, preventing food spoilage.

6. Supply Chain Visibility

- **GPS + IoT Trackers:** Track delivery trucks and shipments in real time from warehouse to store floor.

Workflow: [Products with RFID] → [Smart Shelves + Readers] → [Store Gateway] → [Cloud ERP System] → [Manager Dashboard + Auto-Reorder + Analytics Reports]

4b — Explain the functionalities of the smart grid system with its advantages and disadvantages. [5 marks]

A smart grid is an electricity network enhanced with IoT sensors, communication technology, and data analytics to make power generation, distribution, and consumption intelligent and efficient.

Functionalities:

- **1. Smart Metering (AMI):** Advanced Metering Infrastructure — IoT-enabled electricity meters record consumption in real time and transmit data to the utility automatically. Eliminates manual meter reading.
- **2. Demand Response:** The grid communicates with large consumers (factories, HVAC systems) to reduce consumption during peak demand, balancing the grid and preventing outages.
- **3. Fault Detection & Self-Healing:** Sensors throughout the distribution network detect faults (short circuits, equipment failures) and automatically reroute power to minimize outages.
- **4. Renewable Integration:** Smart grid dynamically manages variable power from solar panels and wind turbines, balancing supply-demand fluctuations in real time.
- **5. Electric Vehicle (EV) Charging Management:** Smart charging stations coordinate EV charging schedules to avoid grid overloads during peak hours.
- **6. Real-Time Monitoring & Analytics:** Operators get real-time visibility into grid health, power quality, and energy flows via centralized SCADA dashboards.

Advantages:

- Reduced energy waste and improved efficiency.
- Faster fault detection and power restoration (self-healing).
- Enables integration of renewable energy sources.
- Empowers consumers with usage data and dynamic pricing.
- Reduces operational costs for utilities.

Disadvantages:

- High initial investment for infrastructure upgrades.
- Cybersecurity vulnerabilities — a hacked smart grid can cause widespread outages.
- Privacy concerns — detailed usage data can reveal personal habits.
- Complex interoperability challenges between legacy and new systems.
- Dependency on reliable internet/communication infrastructure.

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SECTION 3 — Answer Any One

3a — Differentiate the deployment models of IoT level-4 and level-6 with diagram. [5 marks]

IoT Level-4:

- **Architecture:** Multiple sensor nodes → Local gateway → Cloud platform for data storage and analysis.
- **Analysis:** Cloud-based analytics. Data sent from gateway to a cloud service (AWS, Azure, Google Cloud).
- **Storage:** Cloud storage — scalable and accessible globally.
- **Example:** Smart weather monitoring system where sensor data is sent to ThingSpeak cloud for graphing and analysis.

Level-4 Diagram:

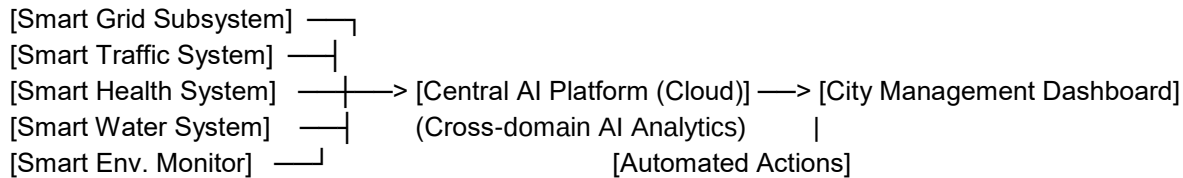
[Sensors] → [Gateway] —Internet—> [Cloud Analytics + Storage] → [User Dashboard]

IoT Level-6 (Ultimate Integration):

- **Architecture:** The most complex level — combines multiple independent IoT Level-1 to Level-5 systems into a fully coordinated, autonomous, centrally managed super-system.
- **Central Coordination:** A centralized intelligent platform (often AI-driven) aggregates data from multiple subsystems (smart grid, smart traffic, smart water, smart health) and orchestrates city-wide or enterprise-wide automation.

- **Analysis:** Multi-layered analysis — edge, fog, and cloud. ML/AI-driven decision making. Cross-domain data fusion.
- **Storage:** Hybrid — local edge nodes + distributed cloud.
- **Example:** A fully integrated smart city platform that simultaneously manages traffic, utilities, emergency services, and environmental monitoring from one AI-driven control center.

Level-6 Diagram:



Feature	Level-4	Level-6
Scope	Single application domain	Multiple integrated domains
Intelligence	Cloud analytics	AI/ML cross-domain intelligence
Automation	Domain-specific	Fully autonomous, city-wide
Complexity	High	Very High
Example	Smart weather station	Integrated smart city platform

3b — Describe different types of IoT communication models with diagrams. [5 marks]

(Complete coverage of all 4 IoT communication models — same as Page 1 Section 3b)

1. Request-Response (HTTP/CoAP)

- Synchronous, client-initiated. Client sends request; server replies. Used for on-demand data fetch.
- Diagram: [IoT App Client] —HTTP GET /temperature—> [IoT Device Server] —{24.5°C}—> [Client]

2. Publish-Subscribe (MQTT)

- Asynchronous, decoupled. Sensor publishes to topic on broker; multiple apps subscribe to receive data.
- Diagram: [Temp Sensor] —pub 'farm/temp'—> [MQTT Broker] —> [Farmer App] [Cloud Logger]

3. Push-Pull (ZeroMQ)

- Producer pushes to queue; multiple consumers pull tasks independently. Enables parallel processing.
- Diagram: [Data Producer] —push—> [Task Queue] <—pull— [Worker 1] [Worker 2] [Worker 3]

4. Exclusive Pair (WebSocket)

- Persistent full-duplex connection. Both sides send messages freely. Low latency real-time communication.
- Diagram: [Control Dashboard] <==== WebSocket (persistent, bidirectional) ====> [Robot/Machine Controller]

SECTION 4 — Answer Any One

4a — Explain the smart parking system using IoT with a neat level of deployment diagram. [5 marks]

A smart parking system uses IoT sensors and cloud connectivity to guide drivers to available parking spots, reducing traffic congestion and fuel waste.

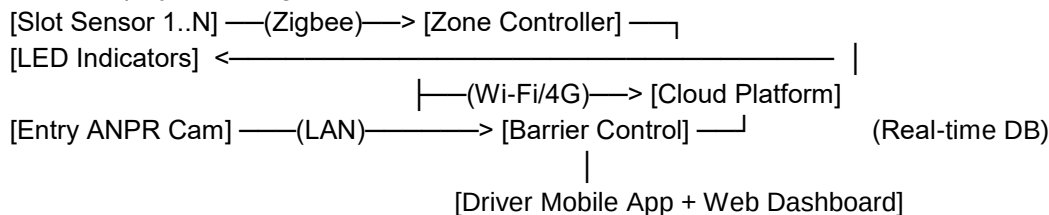
Components & Working:

- **1. Parking Sensors:** Ultrasonic or IR sensors embedded in each parking slot detect whether the slot is occupied or vacant. Status: 0 = vacant, 1 = occupied.
- **2. Slot Status Indicator:** Green/red LED lights above each slot visually indicate availability to drivers inside the parking area.
- **3. Microcontroller Node:** Collects sensor data from a cluster of parking spots and transmits to the gateway via Zigbee or Wi-Fi.

- **4. IoT Gateway:** Aggregates status data from all parking zones, pre-processes it, and sends updates to the cloud platform every few seconds.
- **5. Cloud Platform:** Maintains a real-time map of parking slot availability. Runs reservation and payment APIs.
- **6. Mobile App:** Shows real-time parking map, allows reservation, provides navigation to the selected spot, and handles payment.
- **7. Entry/Exit Barriers:** ANPR (Automatic Number Plate Recognition) cameras or RFID readers manage vehicle entry/exit automatically.

Deployment Level: Level-4

Level-4 Deployment Diagram:



Benefits:

- Reduces time spent searching for parking by up to 30%.
- Decreases urban traffic congestion and fuel consumption.
- Generates revenue data and usage analytics for parking operators.

4b — Discuss the implementation of different IoT devices and their advantages in agriculture. [5 marks]

IoT devices in agriculture enable precision farming — targeted use of resources, real-time monitoring, and data-driven decision making.

IoT Devices & Implementation:

- **1. Soil Moisture Sensors (e.g., FC-28, Decagon 5TM):** Buried in the root zone. Continuously measure volumetric water content. Trigger irrigation only when soil is dry. Advantage: Saves 30-50% irrigation water.
- **2. Weather Stations (e.g., Davis Instruments, DIY LoRa-based):** Measure temperature, humidity, wind, rainfall, UV index. Provide localized microclimate data for spray scheduling and crop protection. Advantage: Enables proactive pest and disease management.
- **3. Drone-mounted Multispectral Cameras:** Capture NDVI images across large fields. Identify stressed or diseased areas early. Advantage: Early intervention reduces crop loss significantly.
- **4. Automated Irrigation Controllers (e.g., Raspberry Pi + solenoid valves):** Schedule and actuate irrigation based on sensor data and weather forecasts. Advantage: Reduces labor, ensures optimal crop hydration.
- **5. Livestock Wearables (GPS + biosensor collars):** Track animal location and health (temperature, heart rate, activity). Early detection of illness. Advantage: Reduces veterinary costs and animal mortality.
- **6. Greenhouse Automation (CO2, light, humidity sensors + actuators):** Automated ventilation, shading, heating/cooling to maintain optimal greenhouse conditions. Advantage: Year-round production, improved yield quality.
- **7. LoRa-based Sensor Networks:** Long-range, low-power communication across large farms where cellular coverage is poor. Connects all field sensors to a central gateway. Advantage: Cost-effective, battery-powered sensor deployment over hectares.

Overall Advantages of IoT in Agriculture:

- Precise resource application → lower input costs (water, fertilizer, pesticide).
- Higher crop yields through optimized growing conditions.
- Remote monitoring → reduced labor requirements.
- Early detection of disease, pests, and equipment faults.
- Sustainability — reduced environmental impact through precision use of chemicals.

SECTION 3 — Answer Any One

3a — Differentiate different IoT level-4 and level-5 with diagram. [5 marks]

IoT Level-4:

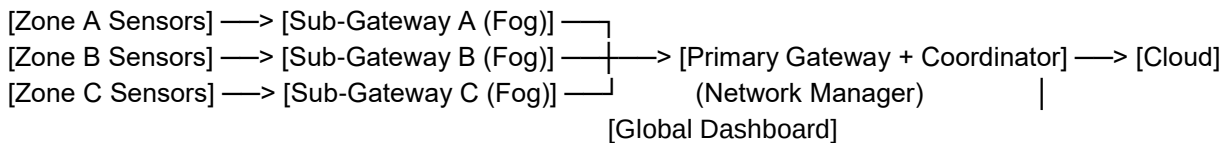
- **Architecture:** Multiple sensor nodes → Local gateway → Cloud platform.
- **Analysis:** Cloud-based — data is sent to the cloud for storage, processing, and visualization.
- **Features:** Remote access via internet, global data visibility, limited edge intelligence.
- **Example:** Smart weather monitoring — sensors report to a Raspberry Pi gateway which pushes data to ThingSpeak cloud.

Level-4: [Nodes] → [Gateway] → [Cloud] → [Dashboard]

IoT Level-5:

- **Architecture:** Multiple sensor nodes → Multiple sub-gateways → Primary gateway → Cloud + Coordinator node.
- **Analysis:** Hierarchical — some analysis at sub-gateways (edge/fog), deeper analysis at cloud. Introduces fog computing layer.
- **Features:** Hierarchical routing, reduced cloud load, lower latency for time-sensitive decisions made at the fog layer, higher fault tolerance.
- **Coordinator Node:** A dedicated network coordinator (e.g., a more powerful gateway) manages the entire IoT network topology.
- **Example:** Smart building management — HVAC, lighting, and security subsystems each have local gateways; a building management server (coordinator) integrates all, with critical data also sent to cloud.

Level-5 Diagram:



Feature	Level-4	Level-5
Analysis	Cloud only	Fog (edge) + Cloud (hierarchical)
Gateways	Single gateway	Multiple sub-gateways + primary
Latency	Higher (cloud round-trip)	Lower (fog handles local decisions)
Complexity	High	Very High
Fault Tolerance	Single gateway failure = system down	Sub-gateway failure = partial impact only
Example	Weather monitoring	Smart building management system

3b — Describe all IoT communication models with diagrams. [5 marks]

Four IoT communication models — covered completely:

1. Request-Response (HTTP / CoAP)

- **Mechanism:** Synchronous. Client requests resource by URI; server responds with data + status code. Connection is short-lived.
- **IoT Example:** Smart meter app sends HTTP GET /energy/usage → meter controller responds with JSON energy consumption data.
- [Client App] —GET request—> [IoT Device/Server] —200 OK + data—> [Client App]

2. Publish-Subscribe (MQTT / AMQP)

- **Mechanism:** Asynchronous. Publishers send to topics on broker; subscribers receive matching topics. Neither knows the other exists.
- **IoT Example:** Farm soil sensor publishes moisture reading to broker topic 'farm/soil/zone1'; irrigation controller and farmer app both subscribe.
- [Sensor Node] —pub 'farm/soil/zone1'—> [MQTT Broker] —> [Irrigation Ctrl] + [Farmer App]

3. Push-Pull (ZeroMQ / Pipeline)

- **Mechanism:** Producers push work items to a queue/pipeline. Multiple consumers pull items for parallel processing. Balances load.
- **IoT Example:** Image frames from security cameras pushed to queue; multiple AI worker nodes pull frames for face recognition processing.
- [Camera Nodes] —push frames—> [Frame Queue] <—pull— [AI Worker 1] [AI Worker 2]

4. Exclusive Pair (WebSocket)

- **Mechanism:** Persistent full-duplex TCP connection. Both client and server can initiate messages at any time. Very low latency.
- **IoT Example:** Industrial control panel maintains a WebSocket connection with a CNC machine — operator sends commands, machine sends status updates simultaneously.
- [Control UI] <==== persistent WebSocket ====> [Machine Controller]

SECTION 4 — Answer Any One

4a — Explain the Application Layer in the IoT protocol suite. [5 marks]

(See Page 2 Section 4b for detailed Application Layer coverage — MQTT, CoAP, HTTP, AMQP, WebSocket, DDS, XMPP with comparison table.)

Summary of Key Points:

- **Position:** Topmost layer in IoT protocol stack. Defines how data is formatted, exchanged, and interpreted between IoT applications and devices.
- **MQTT:** Lightweight pub-sub over TCP. Best for constrained devices. QoS levels 0, 1, 2 for delivery guarantee control.
- **CoAP:** REST-like over UDP. Methods: GET, POST, PUT, DELETE. Supports observe mode for push notifications. Designed for microcontrollers.
- **HTTP/REST:** Heavy but universal. Used where devices have sufficient resources and connectivity.
- **AMQP:** Enterprise messaging — queuing, routing, transactions. Industrial IoT.
- **WebSocket:** Real-time bidirectional. Persistent low-overhead connection after initial HTTP upgrade.
- **Data Formats:** JSON (human-readable, REST/MQTT), CBOR (compact binary, CoAP), XML (legacy systems), Protocol Buffers (high performance).

4b — Write Short Notes on 1) Network Layer 2) Transport Layer 3) Datalink Layer [5 marks]

1. Network Layer in IoT

The Network Layer provides logical addressing and routing of data packets between IoT devices across networks.

- **Key Protocols:** IPv4 (32-bit, limited), IPv6 (128-bit, essential for IoT scale), 6LoWPAN (IPv6 adaptation for low-power networks), RPL (routing for lossy IoT networks).
- **6LoWPAN Role:** Compresses IPv6 headers to fit IEEE 802.15.4 frames, enabling IP connectivity on constrained sensor nodes.
- **RPL:** Builds a DODAG tree for routing in mesh IoT networks with unreliable links and limited resources.
- **Function:** Determines the best path for data from source sensor node to destination cloud server through intermediate routers/gateways.

2. Transport Layer in IoT

The Transport Layer provides end-to-end communication services between IoT devices and cloud applications.

- **TCP:** Connection-oriented, reliable, ordered delivery. Used with MQTT, HTTP. 3-way handshake + flow control + retransmission. Heavy for constrained devices.

- **UDP:** Connectionless, fast, no delivery guarantee. Used with CoAP, DNS, real-time streaming. Low overhead — ideal for IoT sensor telemetry.
- **DTLS:** Datagram TLS — security for UDP in IoT. Provides encryption and authentication for CoAP messages.
- **Function:** Segments data from application layer, manages port-based multiplexing, and ensures appropriate delivery based on application QoS requirements.

3. Data Link Layer in IoT

The Data Link Layer handles node-to-node data transfer within a local network segment and provides MAC addressing and error detection.

- **IEEE 802.15.4:** Foundation standard for low-power, short-range IoT networks. Used by Zigbee, 6LoWPAN. 127-byte max frame. Supports star and mesh topologies.
- **IEEE 802.11 (Wi-Fi):** Higher data rate, higher power consumption. Used in IoT where power is not a constraint.
- **Bluetooth/BLE (802.15.1):** Short-range, low-power. Used in wearables and personal area IoT networks.
- **MAC Protocols:** CSMA/CA (contention-based), TDMA (schedule-based), Hybrid. Manage medium access to prevent collisions in shared wireless channels.
- **Function:** Frames data for physical transmission, performs MAC addressing, error detection (CRC), and controls access to the physical communication medium.

IoT Mid-Term Exam — Sections 3 & 4 Complete Answers (Pages 6–10)

Silicon University | BTCS-T-PC-056 | 5 Marks Each Question

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SECTION 3 — Answer Any One

3a — Describe all IoT communication models with diagrams. [5 marks]

There are four primary IoT communication models, each suited to different data flow requirements:

1. Request-Response Model

- **Description:** The most basic communication pattern. A client sends a request to the server and waits synchronously for a response. Connection is typically short-lived.
- **Protocols:** HTTP (IoT REST APIs), CoAP (for constrained devices).
- **Characteristics:** One-to-one, synchronous, client-initiated, stateless (in REST).
- **Use Case:** Mobile app queries current door lock status. Utility reads energy meter on demand.
- **Diagram:** [Client App] —HTTP GET /status—> [IoT Device Server] —{"status":"locked"}—> [Client]

2. Publish-Subscribe Model

- **Description:** Publishers send messages to a central broker on named topics. Subscribers register interest in topics and receive all matching messages. Publishers and subscribers are fully decoupled — they don't need to know each other.
- **Protocols:** MQTT (most common in IoT), AMQP, DDS.
- **Characteristics:** Asynchronous, one-to-many, event-driven, broker-mediated.
- **Use Case:** A fleet of factory machines publish performance metrics. Multiple analytics and alerting apps subscribe.
- **Diagram:** [Machine 1..N] —pub 'factory/metrics'—> [MQTT Broker] —> [Analytics App] [Alert App] [Logger]

3. Push-Pull Model

- **Description:** Producers push data/tasks into a pipeline or queue. Consumers (workers) independently pull tasks from the queue for processing. Naturally load-balances across multiple consumers.
- **Protocols:** ZeroMQ (PUSH/PULL sockets).
- **Characteristics:** Asynchronous, decoupled, load-balanced, scalable processing.
- **Use Case:** Video frames from security cameras pushed to queue; AI workers pull frames for parallel face detection.
- **Diagram:** [Camera Nodes] —PUSH—> [Frame Queue] <—PULL— [AI Worker 1] [AI Worker 2] [AI Worker 3]

4. Exclusive Pair Model (WebSocket)

- **Description:** A persistent, stateful, full-duplex connection between exactly one client and one server. Once established, either party can send messages at any time without a new request cycle.
- **Protocols:** WebSocket (ws:// or wss://).
- **Characteristics:** Synchronous after setup, bidirectional, low latency, stateful, persistent.
- **Use Case:** Real-time industrial dashboard displaying machine vibration data 100 times/second. Robot remote control application.
- **Diagram:** [Browser Dashboard] <== full-duplex persistent WebSocket ==> [Industrial Controller]

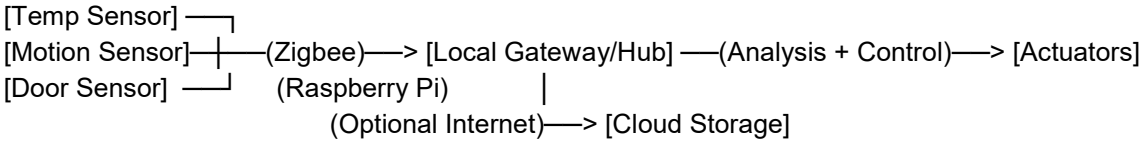
3b — Explain the IoT Deployment Level-3 and Level-4 with a neat diagram. Show how deployment level-4 is different from deployment level-3. [5 marks]

IoT Deployment Level-3:

Level-3 introduces cloud connectivity but keeps analysis at the local gateway level. Multiple nodes send data to a local gateway, which performs analysis and can also forward summarized data to cloud for storage.

- **Key Feature:** Local data analysis at the gateway — real-time decisions are made locally without cloud round-trip latency.
- **Cloud Role:** Optional or used only for long-term storage and remote monitoring, not for real-time decision-making.
- **Connectivity:** Devices use Zigbee/BLE/Wi-Fi to gateway; gateway may have optional internet uplink.
- **Example:** Smart home system — multiple room sensors → Raspberry Pi hub (local analysis, controls AC/lights automatically) → optionally logs to Google Sheets for history.

Level-3 Diagram:

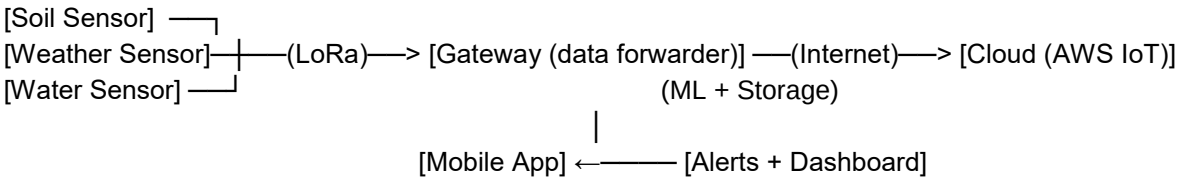


IoT Deployment Level-4:

Level-4 moves the primary analysis to the cloud. The gateway acts as a data forwarder with minimal local processing. Cloud provides scalable storage, deep analytics, machine learning, and global remote access.

- **Key Feature:** Cloud-centric architecture — data flows from devices → gateway → cloud for all analysis.
- **Cloud Role:** Primary analysis engine, long-term storage, ML processing, API delivery to apps.
- **Remote Access:** Globally accessible dashboard and control via internet.
- **Example:** Smart agriculture system — soil/weather sensors → LoRa gateway → AWS IoT → ML models predict irrigation needs → farmer notified via mobile app.

Level-4 Diagram:



Feature	Level-3	Level-4
Primary Analysis	Local gateway (edge)	Cloud platform
Internet	Optional	Mandatory
Real-time Response	Faster (local)	Slower (cloud round-trip)
Remote Access	Limited/Optional	Full global access
Scalability	Limited by local hardware	Highly scalable (cloud)
Example	Smart home hub control	Smart agriculture with ML forecasting

SECTION 4 — Answer Any One

4a — Describe the role of IoT in Agriculture. [5 marks]

IoT is transforming agriculture from traditional manual farming to precision, data-driven farming — enabling higher yields, lower costs, and more sustainable practices.

1. Precision Irrigation

- **Devices:** Soil moisture sensors (FC-28, Decagon), weather stations, automated solenoid valve controllers.
- **Working:** Sensors continuously monitor soil moisture. When moisture drops below a threshold, the IoT controller automatically activates irrigation pumps. Weather forecast integration prevents irrigation before rain.

- **Benefit:** Reduces water consumption by 30–50%, lowers electricity cost, prevents overwatering and root rot.

2. Crop Health Monitoring

- **Devices:** Multispectral cameras on drones, leaf wetness sensors, NIR sensors.
- **Working:** NDVI (Normalized Difference Vegetation Index) maps from drone imagery identify stressed, diseased, or nutrient-deficient crop areas. Early alerts enable targeted treatment.
- **Benefit:** Early disease detection reduces crop loss by up to 40%.

3. Soil Health Management

- **Devices:** Soil NPK sensors, pH sensors, EC (electrical conductivity) sensors.
- **Working:** Real-time soil nutrient data guides fertilizer application — applying the right nutrients in the right amount at the right location.
- **Benefit:** Reduces fertilizer usage by 20–30%, prevents soil degradation.

4. Greenhouse Automation

- **Devices:** CO2 sensors, temperature/humidity sensors, light intensity sensors, automated ventilation/shading actuators.
- **Working:** IoT controllers maintain optimal greenhouse microclimate automatically, adjusting ventilation, heating, and supplemental lighting based on sensor readings.
- **Benefit:** Year-round crop production, improved quality, reduced human labor.

5. Livestock Management

- **Devices:** GPS collars, ear tag biosensors, RFID feeding stations.
- **Working:** Continuously monitor animal location, body temperature, heart rate, and activity. Detect illness early. Track feeding and milk production automatically.
- **Benefit:** Reduced livestock mortality, improved herd health management.

6. Supply Chain & Cold Chain

- **Devices:** Temperature/humidity data loggers in transport vehicles and storage facilities.
- **Working:** Ensure produce is maintained at correct temperatures throughout transport from farm to market.
- **Benefit:** Reduces post-harvest losses, ensures food safety compliance.

Technology Stack:

- Field Sensors → LoRa/Zigbee → Agricultural IoT Gateway → Cloud Platform → Farmer Mobile App/Dashboard.

4b — Describe the use of IoT in smart cities. [5 marks]

(See Page 1 Section 4a for full coverage — identical question.)

A smart city uses IoT to improve quality of life, optimize resources, and enable efficient urban management through real-time data and automation:

- **Smart Traffic:** Adaptive signal control based on vehicle density sensors. Reduces congestion and commute time.
- **Smart Street Lighting:** Motion + ambient light sensors dim/brighten street lamps automatically. Saves 40–60% energy.
- **Smart Waste Management:** Fill-level sensors in bins optimize garbage truck routes. Reduces fuel and labor costs.
- **Smart Water Management:** IoT meters detect leaks, monitor flow, and reduce distribution losses. Smart irrigation in public parks.
- **Air Quality Monitoring:** Distributed sensor nodes measure PM2.5, CO2, NO2. Data alerts authorities and citizens via apps.
- **Smart Parking:** Slot sensors guide drivers to available spots via mobile app. Reduces circling traffic by 30%.
- **Public Safety:** AI-powered cameras, gunshot detectors, and emergency response coordination systems.
- **Smart Energy Grid:** IoT-enabled smart meters and demand response systems optimize electricity distribution.
- **e-Governance:** Citizens access services digitally; city managers receive real-time operational dashboards.

SECTION 3 — Answer Any One

3a — Describe the IoT Deployment Level-1 and Level-2 with clear diagram. Illustrate by taking an IoT application in each case. [5 marks]

IoT Deployment Level-1: Single Node, No Network

The simplest form of IoT deployment where a single device handles all tasks — sensing, processing, local storage, and actuation — entirely on its own without any network connectivity.

- **Architecture:** One device with embedded sensor(s), a microcontroller/processor, and local output (display, actuator, buzzer).
- **Processing:** All data analysis and decision-making happens on the device itself (on-chip or on-board).
- **Storage:** Local storage on SD card, EEPROM, or flash memory.
- **No Connectivity:** No internet, no gateway, no external network required.
- **Application:** Standalone digital thermometer — DHT22 sensor connected to Arduino Uno. Arduino reads temperature every second, displays on 16x2 LCD, and activates a cooling fan relay if temperature exceeds 35°C.

Level-1 Diagram:

[DHT22 Sensor] —analog/digital—> [Arduino Uno] —> [16x2 LCD Display]
(Sense Temp/Humidity) (Process, Decide) —> [Relay → Cooling Fan]
(Store on EEPROM)

IoT Deployment Level-2: Multiple Nodes, Local Network, No Cloud

Multiple sensor nodes connected through a local network to a central master/controller. Data is collected, aggregated, and analyzed locally without internet connectivity.

- **Architecture:** Multiple slave sensor nodes + one master controller communicating via local network (I2C, SPI, UART, Zigbee, RS-485).
- **Processing:** Master node performs analysis and decision-making based on aggregated data from all slaves.
- **Storage:** Local storage at master (SD card, local database like SQLite).
- **No Internet:** Fully self-contained local system.
- **Application:** Multi-zone temperature monitoring system in a hospital — each ward has a DHT22 + Arduino slave. A Raspberry Pi master collects all zone readings via RS-485, stores in local SQLite DB, displays on a 7-inch touchscreen dashboard, and sounds a buzzer if any zone goes critical.

Level-2 Diagram:

[Ward A: DHT22+Arduino] —
[Ward B: DHT22+Arduino] — (RS-485/Zigbee) —> [Raspberry Pi Master]
[Ward C: DHT22+Arduino] —
[Ward D: DHT22+Arduino] —
(Collect, Analyze, Store)
[Local Touchscreen Dashboard]
[Buzzer Alert System]

Feature	Level-1	Level-2
Nodes	Single	Multiple (slaves + master)
Network	None	Local only (RS-485/Zigbee/I2C)
Processing	On-device	Centralized master
Internet	No	No
Storage	On-device	Master node (local DB)

Example	Room thermometer	Hospital multi-zone monitor
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3b — What is S-MAC? Explain the working of S-MAC with a clear diagram. [5 marks]

S-MAC (Sensor MAC) is a contention-based MAC protocol specifically designed for wireless sensor networks (WSNs) to reduce energy consumption, particularly from idle listening.

Key Features of S-MAC:

- **Periodic Sleep-Wake Schedule:** Nodes follow a fixed duty cycle — they alternate between active (listen/transmit) and sleep (radio off) periods. Nodes announce their schedules to neighbors.
- **Virtual Clustering:** Neighboring nodes synchronize their sleep schedules to form virtual clusters, ensuring neighbors are awake at the same time for communication.
- **Collision Avoidance:** Uses CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) during active periods — nodes listen before transmitting.
- **Message Passing:** Long messages are fragmented and sent in one active period using virtual carrier sensing (RTS/CTS/DATA/ACK) to prevent hidden node interference.
- **Adaptive Listening:** A node that overhears a packet destined for its neighbor wakes up briefly at the end of that transmission to receive any messages that may follow.

S-MAC Working Cycle:

Each cycle has two phases: SYNC period (broadcast schedule) and DATA period (transmit/receive data or sleep).

Timeline Diagram:

Node A: |—SYNC—|—ACTIVE (Listen/Tx/Rx)—|—SLEEP—|—SYNC—|...
Node B: |—SYNC—|—ACTIVE (Listen/Tx/Rx)—|—SLEEP—|—SYNC—|...
(Same schedule, synchronized)

ACTIVE period breakdown:

|—RTS—|—CTS—|—DATA ———|—ACK—| ← collision-free using virtual carrier sense

Advantages:

- Significantly reduces idle listening energy — the dominant energy cost in WSNs.
- Scalable to large networks through virtual clustering.
- Achieves 50–80% energy savings compared to always-on radios.

Disadvantages:

- Fixed duty cycle introduces latency — data may wait until next active period.
- Schedule synchronization overhead at startup.

SECTION 4 — Answer Any One

4a — Discuss the various communication models in IoT. [5 marks]

(Four IoT communication models — see Page 6 Section 3a for complete detailed coverage with diagrams.)

Quick Reference Summary:

Model	Protocol	Pattern	Direction	IoT Use Case
Request-Response	HTTP, CoAP	Synchronous	Client → Server → Client	On-demand sensor queries
Publish-Subscribe	MQTT, AMQP	Async, decoupled	Publisher → Broker → Subscribers	Real-time telemetry, smart home
Push-Pull	ZeroMQ	Async, queued	Producer → Queue ← Consumers	Parallel data processing pipelines

Exclusive Pair	WebSocket	Sync, persistent	Bidirectional full-duplex	Real-time dashboards, robot control
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4b — Discuss the various application layer protocols relevant in the context of IoT. [5 marks]

The Application Layer defines how IoT devices and applications exchange data. The key protocols are:

1. MQTT (Message Queue Telemetry Transport)

- **Transport:** TCP/IP. Port 1883 (plain), 8883 (TLS).
- **Pattern:** Publish-Subscribe via a broker (e.g., Mosquitto, AWS IoT Core).
- **QoS Levels:** 0 (at most once), 1 (at least once), 2 (exactly once).
- **Strengths:** Very lightweight (2-byte fixed header), designed for constrained devices and unreliable networks. Supports retained messages and last-will testament.
- **Use Case:** Industrial sensor telemetry, smart home automation, connected vehicles.

2. CoAP (Constrained Application Protocol)

- **Transport:** UDP. Port 5683.
- **Pattern:** REST-like (GET, POST, PUT, DELETE). Supports Observe mode for server-push notifications.
- **Strengths:** Very small message overhead (4-byte header). Designed for microcontrollers and M2M communication. Supports multicast.
- **Use Case:** Smart energy meters, constrained sensor nodes communicating via 6LoWPAN.

3. HTTP / HTTPS

- **Transport:** TCP. Port 80 (HTTP), 443 (HTTPS).
- **Pattern:** Request-Response. RESTful API interactions.
- **Strengths:** Universally supported, mature ecosystem, excellent tooling and debugging support.
- **Weakness:** Heavy overhead — large headers, connection setup cost. Not suitable for constrained devices.
- **Use Case:** IoT device management APIs, cloud-to-cloud integrations.

4. AMQP (Advanced Message Queuing Protocol)

- **Transport:** TCP. Port 5672.
- **Pattern:** Publish-Subscribe with queuing, routing, and transactions.
- **Strengths:** Enterprise-grade reliability, message persistence, complex routing rules. Interoperable across languages and platforms.
- **Use Case:** Industrial IoT, enterprise integration, financial IoT systems.

5. WebSocket

- **Transport:** TCP (upgrades from HTTP). Port 80/443.
- **Pattern:** Full-duplex persistent connection (Exclusive Pair).
- **Strengths:** Real-time bidirectional messaging. After initial handshake, frames have only 2–10 bytes overhead.
- **Use Case:** Live monitoring dashboards, interactive control applications, real-time alerts.

6. DDS (Data Distribution Service)

- **Pattern:** Publish-Subscribe without broker (peer-to-peer). High performance, real-time.
- **Use Case:** Autonomous vehicles, robotics, defense, aerospace IoT where ultra-low latency is critical.

Comparison Table:

Protocol	Transport	Overhead	Pattern	Best For
MQTT	TCP	Very Low	Pub-Sub	Low-power sensors, telemetry
CoAP	UDP	Very Low	Req-Resp	Constrained M2M devices
HTTP	TCP	High	Req-Resp	Non-constrained, web APIs

AMQP	TCP	Medium	Queue/PubSub	Enterprise IoT
WebSocket	TCP	Low (after setup)	Full-duplex	Real-time dashboards
DDS	UDP/TCP	Low	P2P PubSub	Robotics, autonomous systems

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SECTION 3 — Answer Any One

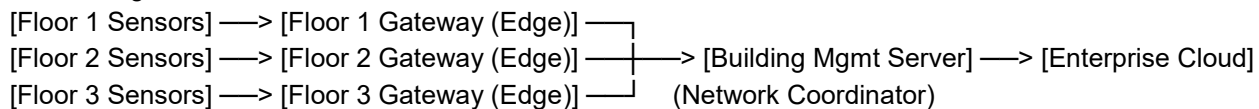
3a — Differentiate different IoT level-5 and level-6 with diagram. [5 marks]

IoT Deployment Level-5:

Level-5 introduces a hierarchical multi-gateway architecture with a network coordinator. It adds a fog/edge computing layer between the sensor nodes and the cloud.

- **Architecture:** Sensor nodes → Multiple sub-gateways (edge/fog nodes) → Primary gateway/coordinator → Cloud.
- **Analysis:** Time-critical processing at edge/fog sub-gateways; long-term analytics at cloud.
- **Coordinator:** A dedicated network management entity controls topology, routing, and inter-gateway coordination.
- **Example:** Smart building — each floor has its own sub-gateway managing HVAC, lighting, and security sensors. A building management server coordinates all floors and reports to the enterprise cloud.

Level-5 Diagram:

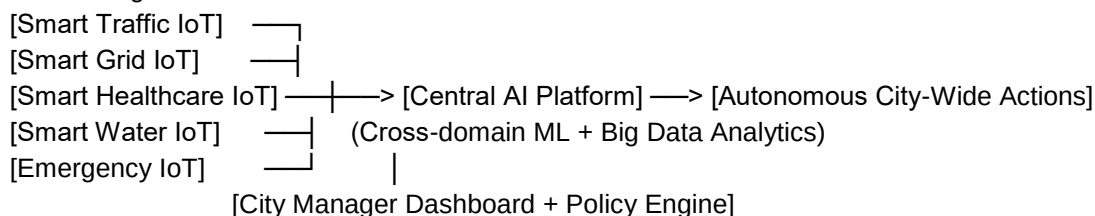


IoT Deployment Level-6 (Ultimate Integration):

Level-6 is the highest and most complex deployment level. It integrates multiple independent IoT systems (possibly Level 1–5) into a single unified, intelligent, autonomous platform. AI/ML drives cross-domain decision-making and automation.

- **Architecture:** Multiple complete IoT subsystems integrated via a central AI-driven coordination platform (cloud + fog + edge).
- **Analysis:** Cross-domain AI/ML fusion — correlates data from multiple application domains for holistic insights and autonomous actions.
- **Autonomy:** Self-configuring, self-healing, self-optimizing across all subsystems.
- **Example:** Smart city platform integrating traffic management, smart grid, healthcare monitoring, waste management, and emergency response into one AI-coordinated control center.

Level-6 Diagram:



Feature	Level-5	Level-6
Scope	Single application, hierarchical	Multiple applications, integrated
Intelligence	Fog + Cloud analytics	AI/ML cross-domain fusion

Coordination	Network coordinator for one system	Central AI platform for all systems
Autonomy	Partial (per domain)	Full autonomous cross-domain control
Example	Smart building management	Smart city platform

3b — Describe all IoT communication models with diagrams. [5 marks]

(Complete coverage — same as Page 6 Section 3a. Four models: Request-Response, Publish-Subscribe, Push-Pull, Exclusive Pair.)

Concise Summary with Diagrams:

- **1. Request-Response:** Synchronous client-server. [App] —GET—> [Device] —data—> [App]. Protocols: HTTP, CoAP.
- **2. Publish-Subscribe:** Async, broker-mediated, one-to-many. [Sensor] —pub—> [Broker] —> [App1][App2]. Protocol: MQTT.
- **3. Push-Pull:** Async queue-based, load-balanced. [Producer] —> [Queue] <— [Workers]. Protocol: ZeroMQ.
- **4. Exclusive Pair:** Full-duplex persistent. [Client] <==WebSocket==> [Server]. Use case: real-time control.

SECTION 4 — Answer Any One

4a — Explain the Application Layer in the IoT protocol suite. [5 marks]

(Detailed coverage — same as Page 2 Section 4b and Page 7 Section 4b. MQTT, CoAP, HTTP, AMQP, WebSocket, DDS — see those sections.)

Key Points:

- **Role:** Topmost IoT layer. Defines protocols for data exchange, message formats, and application-level interactions between devices and cloud services.
- **MQTT:** Lightweight pub-sub over TCP. QoS 0/1/2. Best for constrained devices. Broker-based.
- **CoAP:** Lightweight REST over UDP. 4-byte header. Observe mode for push. Best for microcontrollers.
- **HTTP:** Universal but heavy. Best for non-constrained devices accessing RESTful APIs.
- **AMQP:** Enterprise-grade queuing and routing. Industrial IoT and reliable enterprise integration.
- **WebSocket:** Full-duplex real-time over TCP. Live dashboards and interactive control systems.
- **Data Formats:** JSON (REST/MQTT), CBOR (CoAP — compact binary), Protocol Buffers (high-performance ML pipelines), XML (legacy enterprise).

4b — Discuss the need of managing the IoT Systems. [5 marks]

IoT system management refers to the ongoing administration, monitoring, and maintenance of all IoT devices and infrastructure throughout their lifecycle. It is essential for reliability, security, and scalability.

Why IoT Management is Needed:

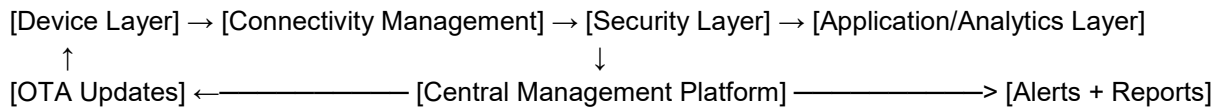
- **1. Scale & Complexity:** IoT deployments can involve thousands to millions of devices spread across large geographic areas. Manual management is impossible — automated management platforms are essential.
- **2. Device Provisioning:** New devices must be securely enrolled, authenticated, and configured before they can join the IoT network. Automated provisioning ensures consistency and prevents unauthorized devices.
- **3. Firmware & Software Updates (OTA):** Security patches and new features must be deployed to all devices remotely. Over-the-Air (OTA) update management ensures devices stay current without physical access.
- **4. Security Management:** Continuous monitoring for anomalous behavior, certificate rotation, access control enforcement, and intrusion detection. A single compromised device can become an attack vector for the entire network.
- **5. Fault Detection & Diagnostics:** Automated health monitoring detects failed sensors, communication outages, battery depletion, and hardware faults. Alerts maintenance teams before issues escalate.

- **6. Performance Monitoring:** Track device uptime, data quality, latency, and throughput. Identify performance degradation trends before they cause failures.
- **7. Configuration Management:** Centrally push configuration changes (sampling rates, alert thresholds, communication parameters) to devices without physical access.
- **8. Data Management:** Manage data retention policies, storage optimization, data quality validation, and compliance with data privacy regulations (GDPR, HIPAA).

IoT Management Platforms:

- AWS IoT Device Management, Azure IoT Hub, Google Cloud IoT Core, Eclipse Hawkbit (open-source OTA).

IoT Management Framework Layers:



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SECTION 3 — Answer Any One

3a — Differentiate the deployment models of IoT level-4 and level-6 with diagram. [5 marks]

(See Page 4 Section 3a for full coverage of Level-4 vs Level-6 with diagrams and comparison table.)

Level-4 Key Points:

- **Architecture:** [Multiple Sensors] → [Gateway] → [Cloud Analysis] → [User Dashboard]. Cloud is the primary analysis engine.
- **Example:** Smart weather station with ThingSpeak cloud dashboard.

Level-6 Key Points:

- **Architecture:** Multiple complete IoT systems (each possibly Level 1–5) integrated into one AI-driven super-platform for cross-domain autonomous management.
- **Example:** Smart city — traffic + grid + water + healthcare all managed by one central AI platform.

Feature	Level-4	Level-6
Domains	Single	Multiple integrated
Intelligence	Cloud analytics	Cross-domain AI/ML
Automation	Domain-specific	City/enterprise wide autonomous
Infrastructure	Gateway + Cloud	Edge + Fog + Cloud + AI
Example	Farm sensor → cloud dashboard	Smart city command center

3b — Describe different types of IoT communication models with diagrams. [5 marks]

(Complete coverage — see Page 6 Section 3a for all 4 models with full descriptions and diagrams.)

Summary:

- **Request-Response:** HTTP/CoAP. Synchronous, client-driven. On-demand data retrieval. Diagram: [Client]—req—>[Server]—resp—>[Client]
- **Publish-Subscribe:** MQTT/AMQP. Async, broker-mediated, one-to-many. Diagram: [Sensor]—pub—>[Broker]—>[Sub1][Sub2]
- **Push-Pull:** ZeroMQ. Queue-based load balancing. Diagram: [Producer]—push—>[Queue]—pull—[Workers]
- **Exclusive Pair:** WebSocket. Full-duplex persistent. Real-time. Diagram: [Client]<==ws==>[Server]

SECTION 4 — Answer Any One

4a — Explain the smart parking system using IoT with a neat level of deployment diagram. [5 marks]

(See Page 4 Section 4a for complete smart parking system coverage with Level-4 deployment diagram.)

Key Components Summary:

- **Ultrasonic/IR Sensors:** One per parking slot — detect occupancy (occupied=1, vacant=0) in real time.
- **LED Indicators:** Red (occupied) / Green (vacant) per slot for visual guidance inside the parking facility.
- **Zone Controller (Arduino/ESP32):** Collects slot sensor data for a cluster of spots and transmits to gateway via Zigbee/Wi-Fi.
- **IoT Gateway (Raspberry Pi):** Aggregates zone data, updates cloud database every few seconds.
- **Cloud Platform:** Real-time slot availability database, reservation system, payment processing.
- **Mobile App:** Shows live parking map, allows slot reservation, provides navigation, processes payment.
- **ANPR Camera:** Automatic entry/exit barrier management and vehicle logging.

Deployment: Level-4 (sensors → gateway → cloud → mobile app).

4b — Discuss the implementation of different IoT devices and their advantages in agriculture. [5 marks]

(See Page 4 Section 4b for complete coverage of all IoT devices in agriculture with advantages.)

Summary of IoT Devices in Agriculture:

IoT Device	Parameter Measured	Key Advantage
Soil Moisture Sensor (FC-28, Decagon)	Volumetric water content	30-50% irrigation water savings
Weather Station (LoRa-based)	Temp, humidity, wind, rain, UV	Localized microclimate, pest management
Drone + Multispectral Camera	NDVI, crop stress, disease maps	Early detection, 40% less crop loss
Soil NPK/pH Sensor	Nutrients, acidity	Precise fertilization, 20-30% cost reduction
Livestock GPS+Bio Collar	Location, temp, heart rate	Early illness detection, reduced mortality
Greenhouse Sensors + Actuators	CO2, temp, humidity, light	Year-round production, labor savings
LoRa Gateway	Long-range data aggregation	Farm-wide coverage on battery power

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SECTION 3 — Answer Any One

3a — Discuss the various link layer protocols that are relevant in the context of IoT. [5 marks]

The Data Link / Link Layer in IoT manages node-to-node data transfer, MAC addressing, and medium access control. Key link layer protocols for IoT are:

1. IEEE 802.15.4

- **Description:** The foundational standard for low-power, short-range wireless IoT communication. Defines PHY and MAC layers for LR-WPANs (Low-Rate Wireless Personal Area Networks).
- **Specifications:** Range: 10–100m, Data Rate: 250 kbps (2.4 GHz), Frame: 127 bytes max.

- **Used By:** Zigbee, 6LoWPAN, Thread. Applications: smart home, industrial monitoring, smart metering.
- **MAC:** CSMA/CA for contention access; TDMA option for guaranteed time slots (beacon-enabled mode).

2. IEEE 802.11 (Wi-Fi)

- **Description:** High-speed wireless LAN standard. Multiple variants: 802.11b/g/n/ac/ax (Wi-Fi 6).
- **Specifications:** Range: 30–100m indoors. Data Rate: 11Mbps (802.11b) to multi-Gbps (Wi-Fi 6).
- **IoT Use:** IoT devices where high bandwidth is needed and power is not a constraint (smart TVs, IP cameras, smart speakers). 802.11ah (Wi-Fi HaLow) targets low-power IoT specifically.

3. Bluetooth / BLE (IEEE 802.15.1)

- **Description:** Short-range wireless technology. BLE (Bluetooth Low Energy) is the IoT-optimized variant.
- **Specifications:** BLE Range: 10–100m, Data Rate: 1–2 Mbps, ultra-low sleep current (< 1µA).
- **IoT Use:** Wearables (fitness bands, smartwatches), health monitors, smart locks, beacons (asset tracking), short-range sensor connectivity.

4. Zigbee (Based on IEEE 802.15.4)

- **Description:** A complete protocol stack built on IEEE 802.15.4. Adds mesh networking, device discovery, and security layers.
- **Specifications:** Range: 10–100m, Data Rate: 250 kbps, supports up to 65,000 nodes per network.
- **IoT Use:** Smart home automation (lighting, thermostats, door sensors), industrial sensor networks, smart metering.

5. Z-Wave

- **Description:** Proprietary low-power mesh networking protocol designed specifically for home automation.
- **Specifications:** Frequency: 868/908/921 MHz (sub-GHz — better wall penetration than 2.4 GHz), Range: ~30m, max 232 nodes/network.
- **IoT Use:** Smart door locks, blinds, thermostats, security sensors in home automation systems.

6. LoRa / LoRaWAN

- **Description:** Long-Range low-power link layer protocol. LoRaWAN is the MAC/network layer built on LoRa PHY.
- **Specifications:** Range: 2–15 km, Data Rate: 0.3–50 kbps, very low power (years on AA battery).
- **IoT Use:** Agriculture (farm-wide sensor networks), smart metering, asset tracking, smart cities — anywhere devices are battery-powered and spread over large areas.

7. NB-IoT / LTE-M (Cellular IoT)

- **Description:** Narrowband IoT — LPWAN (Low Power Wide Area Network) standard using existing cellular infrastructure.
- **Specifications:** Range: up to 10–15 km (cellular), Data Rate: ~26 kbps (NB-IoT), deep indoor penetration.
- **IoT Use:** Smart meters, water/gas meters, parking sensors, logistics tracking where existing cellular coverage is leveraged.

Protocol	Range	Data Rate	Power	Best IoT Use
IEEE 802.15.4	10-100m	250 kbps	Very Low	Foundation for Zigbee/6LoWPAN
Wi-Fi (802.11)	30-100m	Mbps-Gbps	Medium-High	High-bandwidth IoT devices
BLE (802.15.1)	10-100m	1-2 Mbps	Ultra Low	Wearables, health monitors
Zigbee	10-100m	250 kbps	Very Low	Smart home mesh networks
LoRa/LoRaWAN	2-15 km	0.3-50 kbps	Ultra Low	Agriculture, city-wide sensors
NB-IoT	Up to 15 km	~26 kbps	Very Low	Smart metering, tracking

3b — Differentiate different IoT level-3 and level-4 with diagram. [5 marks]

(See Page 6 Section 3b for complete Level-3 vs Level-4 coverage with diagrams and comparison table.)

Quick Summary:

- **Level-3:** [Multiple Sensors] → [Local Gateway with Analysis] → [Local Actuators/Display]. Analysis at edge, no cloud mandatory. Example: Smart home hub.
- **Level-4:** [Multiple Sensors] → [Gateway (data relay)] → [Cloud Analysis + Storage] → [Global Dashboard]. Cloud-centric. Example: Smart agriculture with AWS IoT.

Feature	Level-3	Level-4
Analysis	Local gateway (edge)	Cloud platform
Internet	Optional	Mandatory
Remote Access	No/Limited	Full global access
Real-time Control	Faster (no cloud latency)	Slightly slower
Example	Smart home local control	Farm monitoring with cloud ML

SECTION 4 — Answer Any One

4a — Describe the use of IoT in smart cities. [5 marks]

(See Page 1 Section 4a and Page 6 Section 4b for full smart city IoT coverage.)

Smart City IoT Applications:

- **Smart Traffic Management:** Adaptive traffic signals using vehicle density sensors reduce congestion by 20–25%.
- **Smart Street Lighting:** Motion + LDR sensors control LED street lamps dynamically — 40–60% energy savings.
- **Smart Waste Management:** Fill-level sensors optimize garbage collection routes — reducing fuel costs and emissions.
- **Smart Water Management:** IoT flow meters and leak detectors reduce water distribution losses in aging pipe networks.
- **Air Quality Monitoring:** Dense networks of PM2.5/CO2/NO2 sensors provide real-time pollution mapping for citizen alerts.
- **Smart Parking:** Ultrasonic slot sensors + mobile app navigation reduce parking search time by 30%.
- **Public Safety & Surveillance:** AI cameras + acoustic gunshot detectors + emergency dispatch coordination.
- **Smart Grid:** IoT smart meters + demand response systems optimize electricity distribution and integrate renewables.
- **Healthcare:** Remote patient monitoring, smart ambulances with real-time vitals, hospital capacity management.

Benefits: Reduced operating costs, improved quality of life, lower environmental impact, data-driven governance, enhanced citizen safety.

4b — Describe the use of IoT in smart healthcare. [5 marks]

IoT is revolutionizing healthcare by enabling continuous patient monitoring, remote care, intelligent diagnostics, and efficient hospital management.

1. Remote Patient Monitoring (RPM)

- **Devices:** Wearable ECG patches, pulse oximeters (SpO2), blood pressure monitors, glucose monitors, smart scales.
- **Working:** Wearables continuously collect vital signs and transmit via BLE/Wi-Fi to a healthcare cloud platform. Physicians monitor patient dashboards remotely.
- **Benefit:** Enables chronic disease management (diabetes, hypertension, cardiac conditions) from home — reduces hospital readmissions by up to 30%.

2. Smart Hospital Management

- **Asset Tracking:** RFID/BLE tags on medical equipment (wheelchairs, infusion pumps, defibrillators) enable real-time location tracking — eliminates time wasted searching for equipment.
- **Environmental Monitoring:** Temperature/humidity sensors in operating rooms, medicine storage, and blood banks maintain compliance with strict regulatory standards automatically.
- **Staff & Patient Tracking:** RTLS (Real-Time Location Systems) improve workflow efficiency and emergency response times.

3. Smart Drug Delivery & Management

- **Smart Pill Dispensers:** IoT-enabled dispensers release the correct dose at scheduled times and send compliance alerts to caregivers if medication is missed.
- **Cold Chain Monitoring:** Vaccines and biologics require strict temperature control during storage and transport — IoT sensors ensure cold chain integrity with real-time alerts.

4. Emergency Response

- **Smart Ambulances:** Equipped with IoT medical devices that transmit patient vitals to the hospital ER team before arrival, enabling preparation of treatment.
- **Fall Detection:** Accelerometer-based wearables detect elderly patient falls and automatically call for help.

5. Surgical Robotics & Smart Imaging

- **Connected Surgical Robots:** IoT-connected robotic surgery systems allow specialists to perform minimally invasive surgeries remotely (telesurgery).
- **Smart MRI/CT Systems:** IoT-enabled imaging equipment self-monitors calibration and performance, auto-scheduling maintenance before failures.

6. Mental Health Monitoring

- **Wearable Biosensors:** Monitor stress indicators (heart rate variability, galvanic skin response) and alert care coordinators when intervention may be needed.

IoT Healthcare Architecture:

